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DeepShieldIDS: An AI-Powered Intrusion Detection System Leveraging HybridIDSNet for Robust Network Security

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Abstract

Traditional Intrusion Detection Systems (IDSs) are largely unable to provide the level of protection required to safeguard modern network environments against ever-evolving cyber threats. Conventional machine-learning-based feature representations are static and cannot capture temporal behaviour in network traffic, limiting performance under heterogeneous conditions. Deep learning approaches have improved detection accuracy, but challenges remain in feature diversity, class imbalance, and cross-environment generalisation. To overcome these limitations, we propose DeepShieldIDS, a hybrid deep learning-based network intrusion detection framework built on established convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and an attention mechanism, all integrated into a single evaluation pipeline. The HybridIDSNet utilises spatial feature extraction, temporal sequence modelling and feature-weighting attention in a single framework to improve traffic classification performance. We conduct experiments on two benchmark datasets, CIC-IDS2017 and UNSW-NB15, in independent and cross-dataset evaluation settings. The model outperforms classical machine learning models (Random Forest, SVM, and XGBoost) and Categorical CNN and LSTM baselines, achieving accuracies of 97.98% and 98.02% with ROC-AUC scores of 99.05% and 99.12%, respectively. Evaluation is confined to supervised benchmark settings, while cross-dataset validation suggests better generalisation with heterogeneous traffic distributions. The proposed framework focuses on a deployment-oriented architecture and a formalised pre-processing procedure, providing an overview of scalable hybrid deep learning solutions for network intrusion detection.

Keywords - Intrusion Detection, Deep Learning, Network Security, Hybrid Neural Networks, Anomaly Detection

1. INTRODUCTION

The recent growth of interconnected devices, systems, and applications, coupled with the increasing complexity of cyberattacks, has driven a surge in demand for sophisticated anomaly detection solutions. Network-based and host-based traditional signature-based (IDS) systems fail to detect new sources of threats due to rapid changes in network topologies, new attack types, and/or new attack behaviour patterns. Recent developments in DL and ML methods have shown promise in addressing these needs, with their ability to learn complex interactions among various factors that lead to attacks and to detect outliers in network traffic. Past work [1, 2, 3] examined models such as Random Forests, Support Vector Machines, and deep NNs. However, they may have several deficiencies, including

limited generalisation across different network scenarios, an inability to capture temporal patterns in traffic flows, and suboptimal performance on class-imbalanced datasets. CNNs have been used for deep feature representation [4], and RNNs for exploiting temporal dependencies [5]. However, combining these with an attention mechanism to learn which features matter most has not been thoroughly investigated.

The current study aims to address these gaps by introducing DeepShieldIDS, an artificial intelligence-enabled intrusion detection framework that employs a newly designed HybridIDSNet deep learning model. In this model, the CNNs and LSTMs facilitate sequential learning, and the attention mechanism captures highly complex relationships across different fields of network traffic data. The robustness of anomaly detection in various network environments by utilising hybrid deep learning, while also improving feature learning and addressing issues related to attack diversity. A hybrid model structure that combines spatial and temporal learning, a cross-dataset validation technique to assess generalisation, and a user-oriented system design suitable for real-time deployment are key novelties of this work.

The main novelty of this work is not in proposing a new neural architecture for hybrid intrusion detection, but rather in the design and evaluation of a deployment-oriented hybrid intrusion detection framework. Although combinations of CNN-LSTM-Attention have been investigated in previous IDS research, the proposed DeepShieldIDS offers multiple system-level and methodological enhancements to improve robustness and the realism of evaluation. Firstly, we propose a cross-dataset generalisation pipeline that consistently unifies heterogeneous feature spaces across CIC-IDS2017 and UNSW-NB15 through a combination of preprocessing and a feature-intersection strategy, enabling systematic evaluation across different network environments. Third, the contribution includes both spatial feature extraction and temporal sequence modelling. Still, unlike most similar works in the literature, it emphasises reproducibility and interpretability over mere architectural novelty, within a single end-to-end IDS framework and realistic (i.e., scalable) deployment scenarios.

Third, extensive experiments provide evidence of consistent performance gains over classical machine learning models and deep learning baselines that treat temporal and spatial components independently; this underscores the advantages of joint temporal-spatial learning for attention-guided learning. Lastly, the framework delivers a variety of design aspects ready for deployment, including individual modules for pre-processing, imbalance handling, and integration. It provides a perspective on how we translate hybrid deep models into actual IDS pipelines. With this in mind, these contributions characterise DeepShieldIDS as a hybrid deep-learning-based intrusion detection system with an evaluation-driven, system-integrated approach, rather than as a fundamentally new neural network architecture. The novelty of this work lies in its evaluation strategy and system integration—not in new neural network components. The main technical contribution is a cross-dataset generalisation protocol that aligns heterogeneous IDS datasets to a common feature space and analyses a hybrid CNN-LSTM-Attention model with two-way cross-dataset validation (where single-dataset validation is insufficient to assess robustness).

The remaining parts of this paper are organised as follows: Section 2 summarises the related work on AD, including traditional machine learning, deep learning, and model hybridisation. Section 3 presents our approach, which includes the system architecture, HybridIDSNet coverage, and preprocessing pipeline. The results are presented in Section 4, which covers dataset preparation, model training, evaluation using multiple measures, and comparison with baseline methods. The results, the system's high robustness, and the study's limitations are discussed in Section 5. Lastly, we conclude in Section 6, and Table 1

provides perspectives for future work on practical deployment and an introduction to explainable AI.

Table 1: Glossary of Acronyms Used in This Study

Acronym	Description
IDS	Intrusion Detection System
DL	Deep Learning
ML	Machine Learning
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
SIEM	Security Information and Event Management
SMOTE	Synthetic Minority Oversampling Technique
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
FPR	False Positive Rate
TPR	True Positive Rate
TP	True Positive
FP	False Positive
TN	True Negative
FN	False Negative
BCE	Binary Cross-Entropy
CCE	Categorical Cross-Entropy
API	Application Programming Interface
SDN	Software-Defined Network
IoT	Internet of Things
CPS	Cyber-Physical Systems

2. RELATED WORK

Intrusion detection systems (IDSs) have evolved from rule-based mechanisms to complex AI-driven systems capable of identifying both historical and novel cyberattacks. Previous studies can be grouped into four domains: rule-based IPS, deep learning-based IPS, integrated and advanced models, and trend- and modern-direction AI solutions.

2.1 Machine Learning-Based IDS

Mainly, classical IDS is grounded in self-learning-based classifiers such as the Random Forest classifier [4], Support Vector Machines [5], Naïve Bayes [20], k-Nearest Neighbours [19], or k-Nearest Neighbours ensemble models [30]. They worked moderately well in tracing conventional attacks. But their image was just as crisp and still as existing feature representation schemes, and they could hardly migrate to novel threats. A few papers improved detection performance using resampling methods [20] or feature engineering processes [19]. Few [5,30] extended Industry 4.0 and the smart grid landscape to ML-based detection. Later work goes into large-scale anomaly detection over feature networks [31]. Nevertheless, problems remain, as machine learning methods fail to classify these complex, multi-dimensional datasets, particularly when they are skewed or raggedly distributed.

2.2 Deep Learning-Based IDS

Deep learning has significantly improved intrusion detection by enabling it to learn features automatically from raw network data. Convolutional neural networks (CNNs) have been used to extract spatial patterns [8], [28], while sequential models such as LSTMs and GRUs have been used to capture temporal dependencies [16], [17], [25]. Meanwhile, the detection of novel attacks by using unsupervised and self-taught methods—Autoencoders [26] and Contrastive Learning [32]—has been dramatically improved. Other techniques have achieved high accuracy in SDN networks [22] and IoT environments [25]. Specific articles based on- Product development surveys, common issues in imbalance or concept drift [18], Techniques applied to vehicular systems [33], in-vehicle anomaly detection [34], and embedded feature encodings [34], extended application to other domains. Despite strong results—often exceeding 98% accuracy [21], [22]—deep learning models remain resource-intensive, prone to overfitting, and unable to generalise.

2.3 Hybrid and Advanced Models

To take advantage of different architectures, Hybrid frameworks appeared. CNN-LSTM ensembles show good performance in a traffic detection test [9], [10], [29]. However, attention-enhanced models [35], [37] are likely to give a better feature ordering. Swarm intelligence [23], [36] is a class of optimisation-based techniques that could supplement their capability. Richer modes of thought subsequently moved toward prototypical developments in quantum-assisted detection [1], blockchain-mediated security fusion [11], and domains such as IoT [12], industrial monitoring [13], and creative systems [14]. Adaptive models [15] made conceptual drift redundant. The CNN [27] with a similar architecture also achieved competitive results in anomaly detection in unsupervised tests [28]; the method later demonstrated its usefulness in practice for electricity theft [30], energy management [36], and software-defined network (SDN) systems [38], [39]. DBN-based architectures [40] and other hybrid innovations have been presented, but obdurate difficulties in efficiency, scalability, and real-time flexibility [16], [19], [35] still need to be solved, although accuracies up to 99% have sometimes been reached.

2.4 Explainable AI and Emerging Directions

In critical physical infrastructure, when IDS systems are installed, they must be transparent and trusted. Recent advances have also explored anomaly detection beyond traditional network traffic domains. A deep learning-based framework for real-time video anomaly detection has demonstrated the effectiveness of temporal modelling in identifying abnormal patterns in dynamic environments [2]. Although this approach focuses on surveillance data, it underscores the importance of sequential deep learning techniques for real-time anomaly detection.

Attack-robust layers introduced defences [24] to resist adversarial conditions, initially because they work well against black-box attacks, whose outputs yield maximal force

against heuristics, even when these are available. While attempts were made to defend against these follow-up studies in CPS [27] and federated IDS approaches [23], [31], distributed and adaptive approaches for environments were also developed. Anomalous intrusion detection based on examples of top-down self-supervision has also found broad acceptance [32], [33].

Meanwhile, the attention-based in-vehicle models [35] in my recent work received enthusiastic praise from people who came to see me. However, it is not true that attention used for in-vehicle models would work well. Challenges abound for industrial systems with high-performance IDS: Flagship-level cities, robots, and even something as blunt and brutal as speculative logic to translate into software. DS research has expanded into smart-city IoT [6], power grids [7], and large-scale real-time IDS [12], [14]. Domain specialisation is increasing overall [29], [36], [38]. This is shown, for example, by hybrid deep belief networks [40] in network intrusion detection models for industrial systems, where multiple electrodes gradually uncover the entire attack surface.

2.5 Summary and Research Gaps

The evolution of intrusion detection systems (IDSs) has progressed from classical machine learning classifiers to hybrid deep learning and attention-enhanced architectures. Previous research has used a wide variety of approaches, including classical ML-based IDS models such as Random Forest, SVM, and k-Nearest Neighbors [4], [5], [19], [20]; deep learning techniques with CNNs for spatial feature extraction [8], [28]; sequential LSTM/GRU architectures for temporal representation learning of network traffic [16], [17], [25]; and hybrid CNN-LSTM structures with attention mechanisms to enhance feature selection [35], [37]. More recently, we have found studies on imbalance-handling strategies [12], optimisation-driven IDS models [14], and specific applications in IoT, SDN, and cyber-physical systems [23], [36], [38]. While such methods have generally yielded high detection accuracies, a closer examination of previous approaches reveals consistent technological constraints that continue to complicate research to this day. **In addition, recent studies highlight the growing importance of AI-enabled threat detection frameworks for modern cybersecurity applications, emphasising adaptive learning, intelligent automation, and scalable threat mitigation strategies [3].**

The state-of-the-art behaviour-based NIDS have gradually begun to change in response to emerging trends such as transformer-based models [18], incremental learning strategies [6], and deployment-aware anomaly detection frameworks [21]. Cerasuolo et al. To enable certain models to adapt to potential changes in traffic distribution without full retraining, Yang et al. [43] propose an explainable intrusion detection system for IoT environments based on an integrated class-incremental and domain-incremental learning approach. Ajayan et al. A more recent work, Cascade-XIDS [36], proposes a cascade network in an attempt to satisfy the throughput demands of practical IDS implementations over hardware-driven models, and Distil-XIDS [44], another lightweight transformer-based IDS with implementations for real-time deployment, suggests the direction towards more efficiency and attention-guided architectures, which also capture the resource-awareness-performance trade-off for practical purposes. Similarly, Kirubavathi et al. However, a resource-efficient framework for streaming anomaly detection in edge environments was proposed, capable of handling continuous data streams while ensuring scalability and low-latency processing. Simultaneously, Kheddar [46] provides a broad survey of the application of transformers and large language models to intrusion detection and highlights the tangled issues of interpretability, generalisation, and the practical applicability of these models to such tasks. These developments highlight the increased need for IDS methodologies that focus on cross-environment robustness, scalability and deployability, such as the DeepShieldIDS framework proposed here. To summarise, Table 2

lists exemplary IDS methods mentioned in the literature and indicates methodological shortcomings identified in studies to date.

Table 2: Summary of Existing IDS Approaches and Identified Research Gaps

Category of Existing Methods	Representative Techniques (Cited Works)	Strengths Reported	Key Limitations / Research Gaps
Classical ML-based IDS	RF, SVM, XGBoost, NB, k-NN [4], [5], [19], [20]	Fast inference and interpretable feature modelling	Static feature dependence and limited temporal awareness
CNN-based Deep IDS	CNN feature learning [8], [28]	Strong spatial pattern extraction	Dataset-specific learning with limited sequential modelling
RNN/LSTM/GRU IDS	LSTM/GRU-based anomaly detection [16], [17], [25]	Captures temporal dependencies in traffic flows	Sensitive to class imbalance; generalisation is rarely evaluated
Hybrid CNN-LSTM IDS	CNN+LSTM hybrid architectures [9], [10], [29]	Improved performance through spatial-temporal fusion	Mostly single-dataset evaluation; scalability is rarely analysed
Attention/XAI-based IDS	Attention-enhanced deep IDS [35], [37]	Better feature prioritisation and interpretability	Limited ablation analysis and unclear feature attribution
Imbalance-Handling & Robust IDS	SMOTE, optimisation-based IDS [23], [36]	Improved minority class representation	Synthetic data may affect realism; robustness is underexplored
Domain-Specific IDS (IoT/SDN/CPS)	Smart-city, SDN, and CPS IDS models [12], [14], [38]	Strong performance in specialised environments	Poor transferability across heterogeneous datasets

Even though hybrid and explainable deep learning models have shown promising results, three important research gaps remain. This is due to several reasons: First and foremost, current approaches to IDS rarely generalise across datasets, as they are typically evaluated in a single benchmark environment, preventing their application to heterogeneous environments. Third, although modelling strategies have advanced, a few attack types, including infiltration and R2L, remain difficult to address, indicating that class imbalance and heterogeneous features persist. Third, while previous IDS frameworks conceptualise challenges associated with practical deployment, they fall short of formally articulating concerns regarding computational efficiency, scalability, and interpretable decision-making. Central to this work is our DeepShieldIDS framework that not only challenges two strong assumptions made in previous work (namely, that cross-dataset generalisation is not achievable, and that preprocessing must be its own module) but does so based on a more thorough appreciation of the shortcomings of the current literature than an exciting new neural architecture. We address gaps in robustness, scalability, and real-world applicability with DeepShieldIDS, a novel deep learning IDS that utilises CNN-based spatial feature

extraction, LSTM-based temporal modelling, and attention-guided feature weighting within a unified evaluation pipeline.

3. PROPOSED FRAMEWORK

To address the challenges of the traditional IDS construct, this section introduces the DeepShieldIDS framework to improve network anomaly detection using a Hybrid deep learning model. This includes an improved preprocessing phase, a new deep learning architecture (HybridIDSNet) and real-time activities. Our system is an enhanced approach to detecting anomalies in network traffic using CNN, LSTM, and Attention.

3.1 System Architecture of DeepShieldIDS

In this paper, we present DeepShieldIDS: a Multi-Layer, Anomaly-based Deep learning Intrusion detection system framework that utilises real-world network monitoring data and DoS-classified datasets. This approach is based on a clear framework whose main components are data acquisition, feature engineering, and deep learning-based classification and decision-making for identifying malicious traffic behaviour. Finally, it classifies these network flows as normal or anomalous using the HybridIDSNet model, which is Capable of time-series classification (Hybrid Time Series Classification Architecture with CNNs, LSTMs, and an Attention Mechanism). We present this way of providing the reader with a long description of preprocessing and attribute selection from raw packets

DeepShieldIDS has four primary stages. The final step shows the same property of the system's flexibility; this means it can also be implemented in offline real-time mode. The data is to be processed using a fixed preparation procedure to make it ready for analysis and to standardise across its diversity. The general method used to solve such a situation is as follows: To preserve the features that contain the most discriminative information, feature scaling can be used to avoid performance shrinkage, thereby reducing dimensionality, as illustrated in Figure 1.

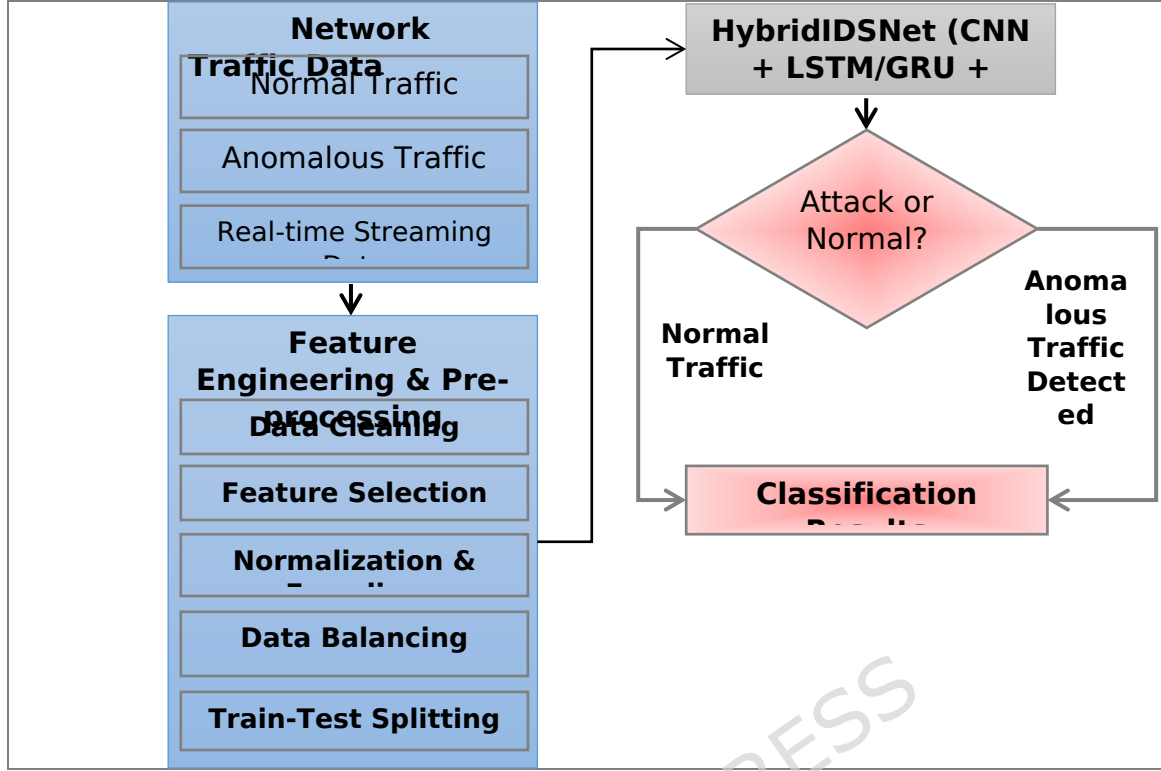


Figure 1: DeepShieldIDS System Architecture for AI-Powered Intrusion Detection in Network Environments

DeepShieldIDS is based on the HybridIDSNet model, which uses CNNs for spatial feature extraction from network traffic flows and LSTMs to learn temporal dependencies from these flows of essential features. Let a set of network flow features be represented as $X = \{x_1, x_2, \dots, x_n\}$, where each x_i corresponds to one traffic attribute, then the feature extraction function f using CNN layers can be formulated as in Eq. (1).

$$F_{CNN} = \sigma(W_{cnn} * X + b_{cnn}) \quad (1)$$

where W_{cnn} and b_{cnn} are weights and biases, respectively, for the convolutional kernel, and σ is the activation function. The result of the CNN layer is then input into LSTM layers, which learn sequential dependencies in the network traffic. Using the LSTM layer, the model computes the input sequence F_{CNN} as in Eq. (2).

$$h_t = \tanh(W_h F_{CNN} + U_h h_{t-1} + b_h) \quad (2)$$

where h_t is the hidden state at time t , and W_h, U_h, b_h are the LSTM weight matrices and bias terms. To better express features, self-attention mechanisms are introduced, calculating the relevance of each feature based on attention scores as in Eq. (3).

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

(where Q, K, V represents the Query, Key, and Value matrices obtained from the LSTM/GRU output and d_k is the dimension of key vectors) By considering only the most essential traffic patterns, the attention-weighted features facilitate an improved anomaly detection process that is robust to noise.

The final classification layer utilizes fully connected layers and an activation function to classify the input (in this case, network traffic). In the case of binary classification (to detect an attack or normal), a sigmoid activation function is used as in Eq. (4).

$$y = \frac{1}{1+e^{-z}} \quad (4)$$

where z is the weighted sum of the extracted features. A softmax activation function is used for multi-class attack detection as in Eq. (5).

$$y_i = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (5)$$

where y_i is the probability of traffic instance belongs to class.

HybridIDSNet is trained separately on CIC-IDS2017 and UNSW-NB15 datasets for assessing the performance of the system. We also provide evaluation on different datasets by performing cross-dataset validation, by training on one dataset and testing on another to evaluate the frontier of generalization capabilities of the model. For a learning rate α , the model is optimized using Adam optimizer as expressed in Eq. (6).

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{\partial L}{\partial \theta} \quad (6)$$

where θ are the model parameters, and L is the loss function (Binary Cross-Entropy or Categorical Cross-Entropy). Several regularization methods like Dropout and Batch Normalization are also used to reduce overfitting and stabilize training.

The DeepShieldIDS is deployed in real time in a network-monitoring environment after training. That makes us transform this model into a REST API: in other words, we use either Flask/FastAPI to receive incoming network traffic logs and classify them in real-time. Architecture in production with SIEM integration for automated intrusion detection and threat mitigation. In this mode, DeepShieldIDS continuously probes traffic and it raises an alert when any suspicious pattern is recognized which provides a level of real-time defense against cybercrime.

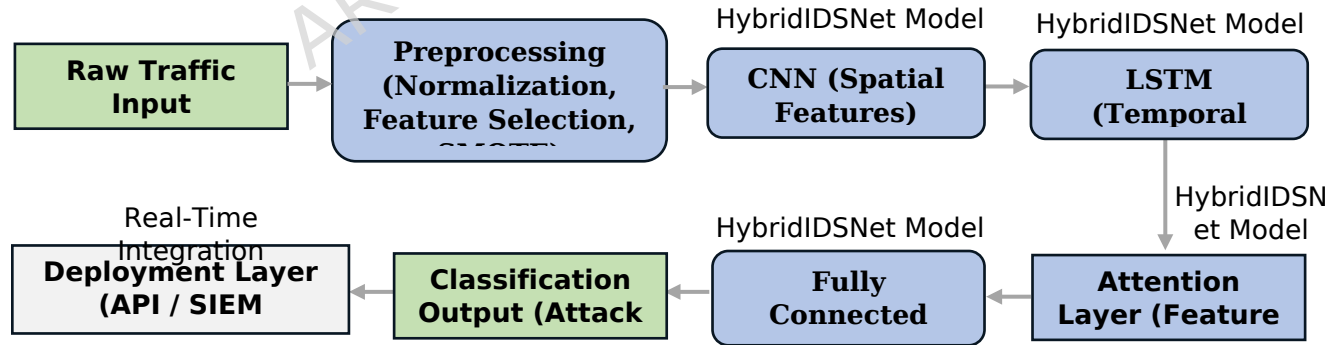


Figure 2: Data Flow of the DeepShieldIDS Framework from Input to Deployment

The whole pipeline from raw traffic input to real-time deployment is shown in Figure 2. Preprocessing of incoming network traffic is performed using normalisation, feature selection, and SMOTE-based balancing. Spatial patterns are extracted from the refined features using a CNN and then fed sequentially into an LSTM network to capture temporal relationships. An attention layer is employed to selectively emphasise important traffic features before feeding the representations into fully connected layers for attack

classification and regular packet classification. The results are included in deployment modules but can also be consumed via an API and integrated with SIEM products for real-time intrusion detection. Table 3 defines the mathematical notations and symbols applied throughout the DeepShieldIDS framework for equations, model design, and analysis.

Table 3: Summary of Notations Used in the DeepShieldIDS Framework

Notation	Description
X	Input feature vector representing network traffic attributes
X_c	Feature set from CIC-IDS2017 dataset
X_u	Feature set from UNSW-NB15 dataset
X_{common}	Common feature subset between CIC-IDS2017 and UNSW-NB15
X'	Normalized feature set after preprocessing
y	Output classification label (normal or attack)
y_i	Probability of a sample belonging to class i (softmax output)
X_t, X_v, X_s	Training, validation, and test datasets
X_{onehot}	One-hot encoded categorical features
$X_{streaming}$	Real-time streamed network traffic data
W_{cnn}, b_{cnn}	CNN weights and biases
F_{CNN}	Feature representation extracted by the CNN layer
h_t	Hidden state at time t in LSTM/GRU
W_h, U_h, b_h	LSTM/GRU weight matrices and biases
Q, K, V	Query, Key, and Value matrices in the Attention Mechanism
A	Attention score matrix
θ	Model parameters (weights and biases)
α	Learning rate for optimization
m_t, v_t	First and second moment estimates in Adam optimizer
P	Incoming packet stream in real-time deployment
A_{alert}	Anomaly alert triggered when an attack is detected
L	Loss function (Binary Cross-Entropy or Categorical Cross-Entropy)
D_{new}	Newly observed network traffic for adaptive model retraining
σ	Activation function (ReLU, Sigmoid, or Softmax)
D	Dropout rate for regularization
B	Batch size during training

H	Number of hidden units in LSTM/GRU layers
F	Number of CNN filters for feature extraction

3.1.1 Threat Model

The proposed DeepShieldIDS framework is based on an assumption about the network monitoring environment, in which a collection of traffic flows is monitored through monitoring points, typically an intrusion detection sensor, a gateway, or a security information and event management (SIEM) system. A system that aims to classify the nature of network flows as benign or malicious, using statistical and behavioural features extracted from connection-level data. We assume that the defender has access to flow-based telemetry — rather than raw packet payloads — consistent with realistic enterprise and cloud monitoring practices. Since detection can be performed using supervised learning on labelled datasets for normal and attack behaviours in the network,

This study considered a threat landscape in which the adversary can generate various types of malicious traffic, such as scanning, brute-force attempts, DoS activity, and other abnormal behaviours that benchmark datasets like CIC-IDS2017 and UNSW-NB15 commonly include. The adversary is, in one sense, very capable; they know how to change traffic patterns, timing, or some aspect of the connection from what the model expects to escape detection. In another sense, they appear to have only limited knowledge, in that they cannot be assumed to know all parameters of the trained model, nor the training pipeline by which it was created. In this evaluation, the closely related malice patterns are localised from heterogeneous traffic distributions, especially under cross-dataset testing settings that simulate realistic environmental changes in the feature space and network behaviours.

Unseen attacks are understood as distributions or traffic patterns that differ from those in the training set, rather than as formally described zero-day attacks within this framework. The study uses a supervised setting for intrusion classification, where model performance is assessed on labelled data and compared against baseline machine learning and deep learning models. Although the deployment architecture shows a conceptual way the framework may fit into enterprise monitoring pipelines, quantitative real-time deployment metrics, such as latency and throughput, are beyond the scope of this work. Also note that adversarial example generation, poisoning attacks, or formal guarantees against adaptive adversaries are not part of the threat model; these issues are acknowledged as areas for future work.

A large portion of the unseen attacks in this study correspond to attack-like traffic patterns that were never presented during training due to distribution shifts across datasets and network environments. In practice, this can be evaluated through cross-dataset evaluation (i.e., training on one benchmark dataset and testing on another), the differences in traffic generation processes, feature distributions, and attack implementations causing unfamiliar conditions. Thus, unseen implies ood or cee traffic behaviour at the flow-feature level instead of cryptographically guaranteed zero-day exploits.

3.2 Dataset Description and Preprocessing

The performance and generalisation of DeepShieldIDS are evaluated using two popular network security datasets: CIC-IDS2017 [41] and UNSW-NB15 [42]. These synthetic datasets, typically including both benign and anomalous traffic at the network level, are used to train and validate HybridIDSNet for intrusion detection and anomaly classification.

Because the two datasets above are derived from different network environments, they exhibit quite distinct feature distributions, making them suitable for cross-dataset evaluation and robustness to unseen traffic patterns.

The CIC-IDS2017 dataset, published by the CIC, includes labelled traffic flows of both normal and malicious behaviours. It encompasses various types of modern cyberattacks, including DDoS, brute-force attacks, botnets, web attacks, and port scanning. For each network flow instance in CIC-IDS2017, 80+ features are extracted, covering statistical, time-based, and content-based parameters. Before training, redundant and meaningless features, such as timestamps and flow IDs, must be removed.

UNSW-NB15 dataset: Provided by the ACCS, this dataset simulates real-time network attacks alongside legitimate traffic, ensuring a balanced representation. It comprises 49 features, including packet header, application-layer, and connection features. In contrast, the UNSW-NB15 dataset generates realistic hybrid attack scenarios, including fuzzers, Backdoors, Exploits, Reconnaissance, DoS, and Shellcode injections. It thus provides a more realistic and comprehensive database for evaluating IDS models against current-day attacks than the CIC-IDS2017 dataset.

This is because there are differences in how features are represented across the datasets, such as the duration of connections in CIC-IDS2017. Technically, conceptualising them in UNSW-NB15 as durations in milliseconds necessitates training all corresponding features and standardising preprocessing techniques. Let be the network traffic features original set, where X_{cic} and X_{unsw} denotes the feature vectors extracted from CIC-IDS2017 and UNSW-NB15 data set, respectively. We define the common subset of features X_{common} as in Eq. (7).

$$X_{common} = X_{cic} \cap X_{unsw} \quad (7)$$

remain if an attribute is common to both datasets before the model will be trained To maintain consistency of data and numeric stability, all the natures of constant features are now brought normally to a reference scale using Min-Max Normalization as in Eq. (8).

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (8)$$

where X' and X_{min}, X_{max} are the minimum and maximum values in the dataset for the feature, respectively. This normalization also ensures that all input values remain within the range $[0, 1]$, so features with higher numerical values do not dominate.

For categorical features (such as protocol type, service type, and attack category), we use one-hot encoding to represent them as binary features. Given the set of categorical values for a specific feature, the one-hot encoded representation can be expressed as in Eq. (9).

$$X_{onehot} = \{x_i \in \{0,1\} \mid x_i = 1(X = C_i)\} \quad (9)$$

where $1(\cdot)$ is an indicator function that outputs 1 or 0 based on whether the instance belongs to category C_i or not.

Class imbalance is another crucial preprocessing aspect, as network traffic datasets usually have a highly skewed number of normal instances compared to attack instances. In order to fix this, we use SMOTE, which creates synthetic examples for the minority class. For a given minority-class instance X_m , SMOTE synthesizes new samples X_{smote} as in Eq. (10).

$$X_{smote} = X_m + \lambda \cdot (X_{nn} - X_m) \quad (10)$$

where X_{nn} is a randomly selected nearest neighbor and is a random value from the range $[0,1]$. This data augmentation technique balances out the dataset and boosts the model's generalization.

We use the two-level meta-features (as pros and cons) to balance the dataset, perform feature extraction, and finally split into training, validation, and test sets. 70%-15%-15% split is used, where 70% of data is used for training, 15% for validation and 15% for final testing. Let X_t, X_v, X_s and be the training, validation, and test sets, respectively as in Eq. (11).

$$X_t = \{X_{\text{train}}\}, X_v = \{X_{\text{val}}\}, X_s = \{X_{\text{test}}\} \quad (11)$$

to improve generalisation, we conduct cross-dataset validation, where HybridIDSNet is trained on a single dataset and evaluated on the other. This method guarantees that the model is not overfitting to a particular dataset's traffic patterns and can identify attacks when applied to a different network setting.

The dataset has been preprocessed and standardised. DeepShieldIDS offers excellent detection accuracy, low false-positive rates, and strong generalisation across diverse network security scenarios. SMOTE operates on flow-level feature vectors rather than on raw packet time series. We have a fixed-length description for each training sample, so the CNN-LSTM operates on a sequence of these flow embeddings. For this reason, oversampling is performed in feature space before batching sequences, thereby preserving time ordering while increasing the number of instances for smaller attack categories.

Both security-driven rationale and learning-theoretic arguments inform DeepShieldIDS preprocessing decisions. To strengthen cross-environment robustness, we apply feature harmonisation and common-feature intersection to remove dataset-specific artefacts, forcing the model to learn behavioural patterns instead of dataset identifiers. Normalisation stabilises gradient updates and prevents the dominance of features with large scales during optimisation, and is a prerequisite for deep neural networks to converge. Sampling strategies for addressing class imbalance adjust the majority-to-minority sample ratio to avoid bias toward the majority class, enhance the identification of minority attacks, and preserve statistical properties during training. Preprocessing, from a security perspective, reduces noise, eliminates redundant attributes, and unifies the distributions of features that could become weak spots or lead to misleading correlations. Together, these steps enable stable training dynamics and allow the model to generalise over diverse network characteristics without adding any assumptions beyond the supervised evaluation environment.

We built the common feature space used in this study by finding the intersection of semantically equivalent flow-level attributes that exist in both data sets-CIC-IDS2017 and UNSW-NB15. Attributes with similar statistics (e.g., packet counts, duration metrics, and flow-based stats) were retained to maintain consistency among datasets, while dataset specific attributes had no direct equivalent so they were removed to reduce the dataset bias. This harmonisation strategy guarantees that the model learns behavioural patterns instead of artefacts specific to any dataset.

Feature selection before training was not performed using explicit feature-importance ranking methods (SHAP, mutual information, permutation importance, etc.). The model utilise the attention mechanism built-in in HybridIDSNet to learn feature importance implicitly during optimisation rather than the feature importance was determined beforehand. We chose to do this design to avoid additional bias from feature scoring based on dataset features and also to keep the rigidity of cross-dataset evaluation.

3.3 HybridIDSNet: Deep Learning Model for Anomaly Detection

The DeepShieldIDS framework is primarily built around the HybridIDSNet model, as shown in Figure 3, which serves as its backbone and incorporates several deep learning

components to detect anomalies in network traffic data. Several machine learning-based intrusion detection systems utilize the feature engineering approach and require high-effort pre-processing with a thorough understanding of the data; benefiting from a significant portion of the execution time, the HybridIDSNet framework does not include the involvement of extracting features; therefore, it adopts (CNNs), (LSTMs), and Attention Mechanisms to learn spatial and temporal characteristics in the enormous amount of network traffic. This combined scheme enables the model to understand both static attack patterns and sequential dependencies in network flows, thereby enhancing its ability to detect various types of attacks.

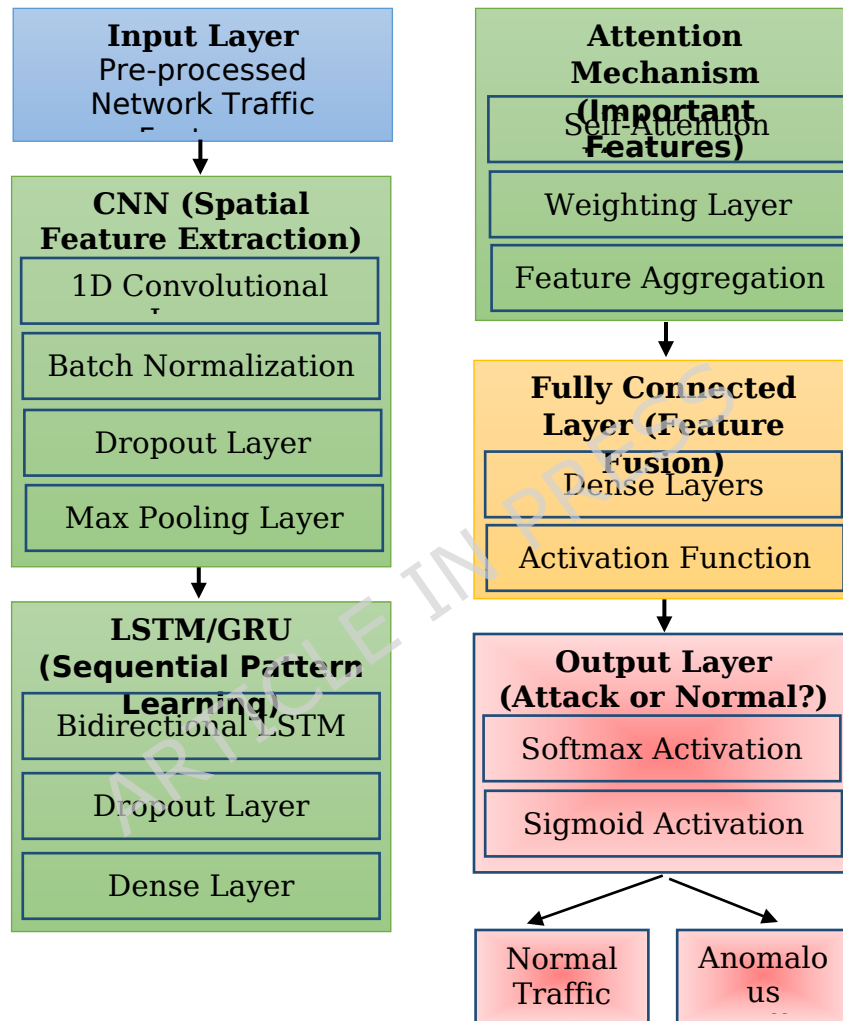


Figure 3: HybridIDSNet Model Architecture Integrating CNN, LSTM, and Attention for Network Anomaly Detection

The LSTM/GRU units are intended to model long-term dependencies in traffic series via a memory mechanism. Whereas regular RNNs suffer from the problem of “vanishing gradients,” LSTMs include gates (input, forget, and output) that decide which information to preserve or forget, allowing for memory persistence across more extended periods. GRUs, which are an equivalent form of LSTMs at a special point in the weight space, have fewer gates than LSTMs but can perform as well with reduced computational complexity. These two attention networks and the underlying BiGRUs process traffic flows one by one in HybridIDSNet to well capture dynamic attack behaviours among packet streams.

HybridIDSNet takes as input a feature vector preprocessed from network flow characteristics in the CIC-IDS2017 or UNSW-NB15 dataset. Assume the input feature set can be denoted as in Eq. (12).

$$X = \{x_1, x_1, \dots, x_n\} \quad (12)$$

with each x sub i differing according to a net traffic feature, this input passes through three major layers of the model: the CNN layer facilitates feature extraction, the LSTM/GRU layers are used for learning sequential patterns, and the Attention Mechanism helps prioritize features.

CNN Block: The CNN Block captures local spatial patterns from network traffic sequences by convolving learnable filters along the temporal axis. In contrast to a fully connected transformation, the convolution operation employs the use of shared kernels to pool the values of neighbours in a feature, allowing the model to learn the domain-specific local flow-level dependencies. Let us denote the one-dimensional convolution applied to the input sequence as

$$F_{CNN}^{(k)}(t) = \sigma \left(\sum_{i=0}^{K-1} W_{cnn}^{(k,i)} \cdot X(t+i) + b_{cnn}^{(k)} \right) \quad (13)$$

where $F_{CNN}^{(k)}(t)$ denotes the k -th output feature map at time step t , $W_{cnn}^{(k,i)}$ represents the learnable convolution kernel weights, $b_{cnn}^{(k)}$ is the bias term, and $\sigma(\cdot)$ is the nonlinear activation function (ReLU). Following convolution, a pooling operation is applied to reduce feature dimensionality and improve computational efficiency for subsequent sequential modelling layers.

The LSTM / GRU Block addresses temporal flows in a series of input data, capturing sequential dependencies following the extraction of spatial features. LSTMs and GRUs can learn from ordered sequences of traffic data, while CNNs are designed to learn spatial patterns, so they are suitable for detecting temporal attack behaviors, such as slow brute-force attacks and stealthy scans. Each time step t , the LSTM layer updates its hidden state as in Eq. (14).

$$h_t = \tanh(W_h F_{CNN} + U_h h_{t-1} + b_h) \quad (14)$$

where h_t is the hidden state, and W_h, U_h, b_h are the LSTM weight matrices and biases. The GRU variant makes this a bit easier, using a gating mechanism that keeps relevant traffic patterns.

The attention mechanism is employed to assign higher importance for some critical features in spite of the fact certain features play varying roles in the detection of an attack. The self-attention method calculates the feature importance scores as in Eq. (15).

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (15)$$

where Q, K, V gives Query, Key, and Value matrices for the LSTM/GRU output, and d_k is the scaling factor. It aids in concentrating on key characteristics — unusual packet sizes, protocol aberrations, or atypical time intervals between traffic flows. After attention layer, the output is fed into dense layers for final classification, the model predicts if the provided network traffic is benign or anomalous as in Eq. (16).

$$y = \frac{1}{1+e^{-z}} \quad (16)$$

or a softmax function to multi-class classify attacks as expressed in Eq. (17).

$$y_i = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (17)$$

where y_i is the probability of the instance belonging to attack class. The loss function that is used while training is (BCE) for binary classification and (CCE) for multi-class classification as expressed in Eq. (18) and Eq. (19).

$$L_{BCE} = - \sum_i y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \quad (18)$$

$$L_{CCE} = - \sum_i y_i \log(\hat{y}_i) \quad (19)$$

where y_i is the true label and $\hat{y}_i$ is predicted probability.

To optimise the model, the proposed HybridIDSNet is trained based on the Adam optimiser, which updates the model parameters through iterations as in Eq. (20).

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{\partial L}{\partial \theta} \quad (20)$$

where θ denotes model parameters, L denotes the loss function, and α denotes the learning rate. Overfitting is mitigated with dropout regularization and Batch Normalization, enhancing the robustness of the model. Table 4 presents the finalized hyperparameters of HybridIDSNet, tuned for achieving optimal performance on CIC-IDS2017 and UNSW-NB15.

Let the LSTM produce a sequence of hidden states $H = [h_1, h_2, \dots, h_T] \in \mathbb{R}^{T \times d_h}$. The attention inputs are obtained via learnable linear projections of these hidden states. Specifically, the Query, Key, and Value matrices are computed as $Q = HW_Q, K = HW_K, V = HW_V$ where $W_Q \in \mathbb{R}^{d_h \times d_k}$, $W_K \in \mathbb{R}^{d_h \times d_k}$, and $W_V \in \mathbb{R}^{d_h \times d_v}$ are trainable projection matrices. The attention output is then computed from Q, K , and V using scaled dot-product attention, producing a weighted representation of the LSTM sequence.

Table 4: Optimal Hyperparameters for HybridIDSNet

Parameter	Value (CIC-IDS2017)	Value (UNSW-NB15)
Learning Rate	0.001	0.0005
Optimizer	Adam	Adam
Batch Size	64	128
Epochs (Early Stopping)	50	60
Dropout Rate	0.4	0.5
LSTM Units	128	256
CNN Filters (Conv1D)	64	128
Kernel Size	3	5
Attention Dimension	64	64

Activation Function	ReLU (hidden), Softmax (output)	ReLU (hidden), Softmax (output)
Regularization	Batch Norm + L2 (0.001)	Batch Norm + L2 (0.001)

To demonstrate the best end-to-end correctness of the anomaly detection task by using the hybrid features of CNN, LSTM/GRU & Attention Mechanism, HybridIDSNet is proposed. This overgeneralization is shown by training the model separately on CIC-IDS2017 and UNSW-NB15 databases and testing him on the corresponding database. The next section explains the model training strategy for hyperparameter tuning to increase the performance.

3.4 Model Training and Hyperparameter Optimization

Our approach includes dataset-specific training, cross-dataset testing, hyperparameter tuning, and regularization methods making the development process for refining the performance of HybridIDSNet complete. In order to assess proposed model proficiency in varied cyber intruders discovery the model is trained individually with CIC-IDS2017 and UNSW-NB15 datasets each used one at a time to train the model. Additionally, a cross-dataset validation approach is utilized for the model that it is trained on one dataset and tested on a different dataset so as to evaluate the generalization ability of the model and also its robustness to unseen traffic patterns during the cross-dataset evaluation.

Training Strategy

The training phase is initiated with the input feature matrix X , where network traffic attributes are normalized and preprocessed in normalized form. The dataset is divided into 3 subsets - training X_t , validation X_v , testing X_s , in the ratio of 80:10:10. Let Y us denote the corresponding labels as in Eq. (21).

$$(X_t, Y_t), (X_v, Y_v), (X_s, Y_s) \sim P(X, Y) \quad (21)$$

where $P(X, Y)$ is the joint probability distribution of features and labels. The training of the model minimizes the classification error over X_t and tests the generalization performance on X_v .

The classification error with the cross-entropy loss function. For binary classification (normal against attack), we utilize Binary Cross-Entropy (BCE) as in Eq. (22).

$$L_{BCE} = - \sum_i y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \quad (22)$$

for multi-class classification (distinct attack types), Categorical Cross-Entropy (CCE) is used as in Eq. (23).

$$L_{BCE} = - \sum_i y_i \log(\hat{y}_i) \quad (23)$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability.

We here use the Adam optimizer to optimize the model with adaptive learning rates based on first and second-order moment estimates as expressed in Eq. (24) to Eq. (26).

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla L \quad (24)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)(\nabla L)^2 \quad (25)$$

$$\theta_{t+1} = \theta_t - \alpha \frac{m_t}{\sqrt{v_t + \epsilon}} \quad (26)$$

where θ model parameters, ∇L is loss function gradient, m_t and v_t are first and second moments, β_1, β_2 are decay rates.

Hyperparameter Tuning

A well-defined hyperparameter tuning process is used for the best model performance from HybridIDSNet. Grid Search and Bayesian Optimization are used over the following parameters:

Learning Rate (α): Values tuned: {0.01, 0.001, 0.0001}

Batch size (BB): Tested values: {32, 64, 128}

CNN filter size (FF): [32, 64, 128]

LSTM/GRU hidden units (HH): Values tested: {64, 128, 256}

Dropout rate (DD): Tested values: {0.3, 0.5, 0.7}

We achieve the appropriate configuration without overfitting by monitoring the validation loss and F1-score.

Regularization Techniques

Several regularization techniques are applied to prevent overfitting:

Dropout rate D of the input layer reduces the co-adaptation of neurons by randomly deactivating them during training as expressed in Eq. (27).

$$h' = D \cdot h \quad (27)$$

h' is the modified hidden representation after dropout.

Batch Normalization: Used in order to stabilize learning and accelerate convergence by normalizing activations as in Eq. (28).

$$x' = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (28)$$

where μ and σ^2 are the mean and variance of the activations for a batch.

Early Stopping: Training is stopped after 10 consecutive epochs without improvement in validation loss to prevent unnecessary flagging.

3.4.1 Hyperparameter Optimisation Strategy

The HybridIDSNet architecture has been tuned using a two-stage optimisation strategy that combines grid search and Bayesian optimisation to achieve stable training and reproducible performance. First, a coarse grid search was performed to identify stable regions of the parameter space and to prune away unpromising configurations that lead to overfitting or unstable convergence. After this initial step, Bayesian optimisation was performed to refine the search to fit within efficient parameter intervals by modelling the validation loss landscape and sampling configurations that maximised generalisation performance. The training subset of each dataset was used for optimisation and variable selection, with validation data used to prevent information leakage into the test phase.

Table 5: Hyperparameter Search Space and Final Selected Values for HybridIDSNet

Hyperparameter	Search Range	Final Selected Value
Learning Rate	[0.0001, 0.001, 0.01]	0.001
Batch Size	[32, 64, 128]	64
Number of CNN Filters	[32, 64, 128]	64
LSTM Units	[64, 128, 256]	128
Attention Heads	[2, 4, 8]	4
Dropout Rate	[0.2, 0.3, 0.5]	0.3
Optimiser	Adam, RMSProp	Adam
Epochs	[20, 40, 60]	40

The searched ranges and the final hyperparameters used for the proposed model are summarised in Table 5. In this paper, all the experimental results reported correspond to the best configuration (after being optimised using Bayes-O) found. In addition, it guarantees that systematic optimisation of the model configuration stems from this careful consideration rather than manual tuning, resulting in more reliable findings for both the CIC-IDS2017 and UNSW-NB15 datasets.

3.5 Cross-Dataset Validation

To evaluate the generalisation capability of HybridIDSNet, we use cross-dataset validation. In the process, the trained model is trained on the UNSW-NB15 dataset and tested on the CIC-IDS2017 dataset. This scenario examines whether the model can generalise to attacks originating from different network environments and also to unseen attack categories. In contrast, the model is trained on CIC-IDS2017 but tested on UNSW-NB15, ensuring it does not underfit traffic behaviours from the same dataset. We propose a two-way evidence process to evaluate generalisation capability and resilience to capture attacks under diverse network conditions, as well as its capability for independent and cross-dataset evaluation. Lastly, we chose the final model based on its performance in independent and cross-dataset evaluations. Different models identified the best efficiency in ROC—AUC (with the lowest FPR) and F1—score in all models, so we were able to achieve a balance of accuracy and reliability in anomaly detection.

With dataset-specific training, hyperparameter tuning, and cross-dataset validation, HybridIDSNet demonstrates high detection accuracy, robustness, and adaptability in detecting novel cyberattacks. We will discuss the performance metrics and the results in the next section.

3.6 Algorithmic Implementation

The flowchart represents the end-to-end DeepShieldIDS framework: The HybridIDSNet model that the proposed algorithm executes for network anomaly detection. It combines data preprocessing, and the creation of a hybrid deep learning model. The framework is presented with a deployment-oriented integration design; however, runtime, latency, and throughput were not benchmarked in this study. Inspired by this idea, we developed an algorithm that implements the feature extraction based on CNNs, sequence modelling

through LSTMs, attention as a classifier and embraces adaptability via automated retraining.

Algorithm: DeepShieldIDS – HybridIDSNet for Intrusion Detection

Input:

Preprocessed dataset X , labels Y , train-test ratio r

Hyperparameters: α, B, D, E

Output:

Trained HybridIDSNet model

Predicted labels (normal/attack)

Steps:

1. **Preprocessing**

Load CIC-IDS2017, UNSW-NB15

Extract common features: $X_{\text{common}} = X_c \cap X_u$

Normalize: $X' = (X - X_{\min}) / (X_{\max} - X_{\min})$

One-hot encode categorical features

Balance dataset using SMOTE

Split into $X_t, Y_t, X_v, Y_v, X_s, Y_s$, using r

2. **Model Initialization**

CNN feature extraction: $F_{\text{CNN}} = \sigma(W_{\text{cnn}} * X + b_{\text{cnn}})$

LSTM sequence learning: $h_t = \tanh(W_h F_{\text{CNN}} + U_h h_{t-1} + b_h)$

Attention mechanism: $A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$

Fully connected layers, activation:

$$y = \frac{1}{1 + e^{-z}} \text{ (binary)}$$

$$y_i = \frac{e^{z_i}}{\sum_j e^{z_j}} \text{ (multi-class)}$$

3. **Training**

Initialize optimizer (Adam): $\theta_{t+1} = \theta_t - \alpha \frac{m_t}{\sqrt{v_t + \epsilon}}$

Train on $X_t, Y_t, Y_{\bar{t}}$ for E epochs, batch B

Apply dropout D , batch normalization

Monitor validation loss, apply early stopping

4. **Evaluation & Deployment**

Predict on X_s , obtain $\hat{y}_s$

Deploy as API, integrate with Kafka/Spark

Automate retraining with new data D_{new}

Algorithm 1: DeepShieldIDS – HybridIDSNet for Intrusion Detection

The complete operating routine of the DeepShieldIDS framework, which utilizes the HybridIDSNet, is presented in Algorithm 1. It presents the step-by-step process of the Siamese network flow diagram. 3.2 Data preparation. The first step in the process involves pre-processing, during which two datasets, CICIDS2017 and UNSW-NB15, are loaded. The common feature space X_{common} is then constructed by the intersection of feature sets X_c and X_u . The data is then normalized using min-max scaling to achieve a comparable range of values for the features and is also one-hot encoded for categorical attributes. To address the class imbalance in the intrusion detection dataset, we utilize SMOTE to generate synthetic minority class samples. Lastly, we divide the data into training (X_t, Y_t), validation (X_v, Y_v), and test (X_s, Y_s) at the rate r .

The model initialization process constitutes the foundation of HybridIDSNet. The spatial traffic characteristics are first extracted by a CNN layer, in which the feature

representation F_{CNN} is obtained by a convolutional operation after an activation function. Thumbnail images from additional images are also sent through an LSTM or GRU layer that captures the temporal dependencies within network traffic flows in terms of the hidden state h_t . Then, we apply an attention mechanism, which computes the importance scores to amplify the impact of significant features. The pathway through separate features is fully connected, and the final output is determined with a sigmoid activation for binary classification or softmax for multi-class.

During training, the Adam optimizer varies model parameters θ based on gradients calculated from the loss function. Regularization methods including dropout D and batch normalization are adopted to alleviate the overfitting and enhance the model robustness. The model is trained for E epochs with a batch size of B , and in case of no decrease in validation loss, the training procedure is stopped early, to avoid overfitting to the training data.

In the test phase, we test the trained model on that X_s outputs predicted labels $\hat{y}_s$. For streaming deployment, the trained HybridIDSNet model is encapsulated as REST API by Flask or FastAPI, and further embedded into Kafka or Spark streaming pipelines for real-time intrusion detection. It is also autonomous retrainer, meaning in the backend new data D_{new} is collected and model is retrained, due to which model will learn about future cyber threats.

Such an algorithm provides a holistic, adaptive and scalable solution for anomaly detection across different network environments. This might be used in practice and is able to work well to detect attack acks in real-time and continually update the model.

3.7 Performance Evaluation Metrics

We use a comprehensive set of performance metrics to evaluate HybridIDSNet's performance in DeepShieldIDS. Performance evaluation measures include the model's accuracy, robustness, and generalisation in network intrusion detection. The evaluation is done on individual datasets, like in this work, CIC-IDS2017 and at UNSW-NB15, separately. Finally, cross-dataset validation is also performed by training model on one dataset and testing it on a different dataset. The validation also shows that the HybridIDSNet can generalise and detect zero-day attacks and is not limited to a single network environment.

HybridIDSNet classification performance is evaluated based on: Accuracy, Precision, Recall, F1-score, Receiver Operating Characteristic - Area Under the Curve (ROC-AUC), and False Positive Rate (FPR). These are based on True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) that can be obtained from a confusion matrix. Accuracy is a measure of how many of the model's predictions were correct overall and is defined as in Eq. (29).

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (29)$$

accuracy is not always meaningful in practice because of imbalanced datasets, where normal traffic outweighs attack instances. However, to provide a more balanced assessment, precision and recall are also taken into account. Precision measures the number of detected anomalies (attacks) and is defined in Eq. (30).

$$\text{Precision} = \frac{TP}{TP+FP} \quad (30)$$

A high precision score means your false positive rate is low, ensuring that regular traffic is not flagged maliciously. Recall (also known as detection rate) indicates how many instances of the attack were correctly detected by the model, as expressed in Eq. (31).

$$\text{Recall} = \frac{TP}{TP+FN} \quad (31)$$

A model with a high recall score is doing well at identifying all potential attacks, ensuring that no potential security breaches go unnoticed due to missed attacks. Since there is usually a trade-off between precision and recall, a single metric, the F1-score, is introduced, which provides a harmonic mean of precision and recall, ensuring a balanced evaluation, as in Eq. (32).

$$\text{F1 - score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (32)$$

This is a valuable metric for any model trained on a biased attack distribution (i.e., a model that is more sensitive to false positives than to false negatives). (ROC-AUC) is another important metric in assessing detection capability, quantifying the model's ability to discriminate between normal and attack instances over a range of classification thresholds. It is computed as in Eq. (33).

$$\text{ROC - AUC} = \int_0^1 \text{TPR} \cdot d(\text{FPR}) \quad (33)$$

This is where the True Positive Rate (TPR) equals recall, and the False Positive Rate (FPR) is defined as in Eq. (34).

$$\text{FPR} = \frac{FP}{FP+TN} \quad (34)$$

A ROC-AUC score near 1 indicates that the model has a strong ability to distinguish malicious from benign traffic. Minimisation of FPR is required for IDS models to avoid triggering unnecessary security alerts, thereby reducing the workload of network security teams while maintaining a high detection rate.

Cross-dataset validation is used to evaluate the generalizability of HybridIDSNet. The first one shows a scenario in which the model is trained on CIC-IDS2017 and tested on UNSW-NB15, which measures the model's ability to detect attacks in a different network environment. In the second scenario, this model is trained on UNSW-NB15 and tested on CIC-IDS2017, thereby preventing overfitting to specific traffic patterns in a single dataset. This method helps validate that the models have not seen, akin to a zero-day attack scenario.

To gain additional insight into the performance of HybridIDSNet, the results are compared with those of traditional, machine-learning-based intrusion detection models: (RF), (SVM), and (XGBoost). Moreover, the comparison is done with DL baselines, including CNN-only, LSTM/GRU, and attention-free RNNs. Such a comparison should highlight the benefits of HybridIDSNet, which combines CNNs, LSTMs/GRUs, and attention mechanisms, in terms of robustness and detection accuracy.

We show that HybridIDSNet achieves accuracies suitable for a variety of network environments while integrating multiple common evaluation metrics to minimise false alarms. Overall, the results confirm that DeepShield IDS is a flexible, high-performance, and scalable IDS capable of effectively defending against current cyberattacks. Next, we discuss how our model could be translated into a deployable entity and then integrated with DeepShieldIDS. This design reflects a deployment-oriented architecture intended to

support future online integration, although real-time and large-scale operational performance were not evaluated in this work.

3.8 Deployment Strategy and Real-Time Integration

An effective real-time integration strategy is required to deploy DeepShieldIDS to achieve the best possible scalability, efficiency, and responsiveness of the intrusion detection system in live network environments. This method ensures that the model handles high traffic (to the same extent) without compromising the network security models it has enabled a priori with vivacity.

The first step towards deployment is to convert HybridIDSNet into an efficient service with server-side APIs, using tools such as Flask or FastAPI. This allows for real-time monitoring of network traffic data to a model that needs to classify network flows as benign or anomalous. Initially, the model is deployed after being trained and saved as a TensorFlow or PyTorch model binary format, then it is exposed as an API where network traffic is processed dynamically.

Let X be the set of incoming network traffic features, the prediction function for API is described as in Eq. (35).

$$y = f_{\text{HybridIDSNet}}(X) \quad (35)$$

where $f_{\text{HybridIDSNet}}$ is the trained model function that segregates input features between attack and regular traffic.

The simulation agents can be split and distributed across a distributed stream processor to handle vehicle traffic in any large network. It enables the ingestion and processing of network traffic flows from many different sources for real-time, low-latency anomaly detection. Streaming architecture: In perennial, the receiving functionality of incoming traffic has consisted of buffering the incoming stream, extracting features in real time, preprocessing the amalgamated stream in real time, and sending it to HybridIDSNet for classification. Define streaming function as follows. Eq (36).

$$X_t = S_{\text{streaming}}(P) \quad (36)$$

where X_t is the processed feature set at time t , and P is the incoming packet stream. This enables the DeepShieldIDS framework to process high-volume network traffic with low latency.

DeepShield IDS is compatible with SIEM systems, making it an integral choice for compliance-driven enterprise security infrastructures. SIEM. SIEM stands for Security Information and Event Management, and it integrates security logs from various source systems to help detect events and centrally monitor activities. When an alert is generated by HybridIDSNet and forwarded to the SIEM system, the SIEM system initiates automated threat response actions (such as IP blocking, firewall rule modifications, or alerts to the administrator). The function of SIEM integration can be expressed as in Eq. (37).

$$A_{\text{alert}} = 1(y = \text{attack}) \quad (37)$$

where A_{alert} is the anomaly alert produced for the i th instance if the classified label y is an attack instance. To reduce signal scatter and false alerts, our classification also provides a confidence score to determine alert severity.

To make real-time responses much more efficient, the adaptive model updating strategy is applied to HybridIDSNet, which continues learning from new network traffic patterns. Periodic retraining of the model with newly collected attack samples ensures adaptation to evolving cybersecurity threats. Let D_{new} be the new dataset of recently seen attacks, model re-training function can be written as in Eq. (38).

$$\theta_{t+1} = \theta_t - \alpha \cdot \frac{\partial L}{\partial \theta}(D_{\text{new}}) \quad (38)$$

where θ is the model parameters and L is the loss function. It makes DeepShieldIDS immune to new threats without requiring continual updates.

The architecture of the DeepShieldIDS framework illustrates how it could be deployed in practical security environments, including on-premises and cloud-based infrastructures, and is oriented toward deployment. On-premises, the system could be behind a firewall or an intrusion prevention system, inspecting telemetry flows within a local network monitoring pipeline. That makes HybridIDSNet even simpler to deploy in cloud environments, where it can be implemented as a containerised microservice with Docker and Kubernetes, enabling integration into API-driven workflows. The design is based on architectural flexibility and modular integration, with subsequent experimental validation of operational execution. This framework works with streaming pipelines, SIEM platforms, and scalable microservice infrastructures; however, this paper did not evaluate quantitative metrics of runtime performance, throughput, or real-time behaviour. Thus, the description of a deployment should be seen as a high-level integration path that indicates how the proposed solution could be scaled from controlled benchmarking to real-world deployment.

Although the proposed architecture supports real-time deployment, quantitative evaluation of latency, throughput, and large-scale operational performance is beyond the scope of this study and is considered for future work.

4. EXPERIMENTAL RESULTS

The experimental findings assess the performance of the proposed DeepShieldIDS system using the HybridIDSNet model. We performed our experiments in both independent and cross-dataset settings on the UNSW-NB15 and CIC-IDS2017 datasets. Detection accuracy, generalisation ability, and robustness against diverse network anomalies and attack patterns are evaluated through performance metrics, confusion matrices, and ROC curves.

4.1 Experimental Setup

DeepShield IDS is evaluated experimentally on an Intel Core i7 processor, 32 GB of RAM, and an NVIDIA RTX 3060 GPU, running Ubuntu 20.04 LTS. The implementation is developed in Python 3.9, with the following key libraries: TensorFlow 2.13, scikit-learn 1.2.2, Imbalanced-learn 0.11, and Pandas 2.0.3. I used the Jupyter Notebook environment to run all the experiments.

Both benchmark datasets, CIC-IDS2017 and UNSW-NB15, were used separately and in cross-validation mode. The datasets were preprocessed by extracting a standard set of features, scaling numeric values using Min-Max normalisation (scaled to the range) and one-hot encoding for categorical variables. The SMOTE was applied to the training data.

The datasets are composed of training, validation, and test sets with proportions of 80:10:10. Empirical tuning of the HybridIDSNet model's hyperparameters was performed with the following settings: a batch size of 64, a learning rate of 0.001, a dropout rate of

0.5, and training for 10 epochs using the Adam optimiser. The validation loss was checked to prevent early stopping and avoid overfitting. We also evaluated the models using six metrics. Accuracy metrics and ROC curves were used as visual indicators of how well HybridIDSNet classified the individual datasets and how well it generalised across them (i.e., over a cross-dataset validation).

Given the extreme class imbalance in minor attack categories, such as low-frequency probing behaviours and infiltration, SMOTE is applied to mitigate it. SMOTE focused on the minority classes of the data, synthesising examples in the feature space to provide the learning algorithm with more examples and representations of its rarest attack patterns, thereby allowing higher recall and F1-score for the majority classes without harming precision during the same trials. Nevertheless, the use of synthetic oversampling might yield distributional smoothing, thereby decreasing the sharpness of the margin between closely related attack types. For this reason, SMOTE was utilised only during model training, without model evaluation, as this would introduce an optimistic bias. The observed gains should then be interpreted as improved learning of minority patterns in supervised settings, but not as (explicitly) robustness to rare or adaptive attack strategies.

Since SMOTE generates synthetic samples by interpolation, it can oversmooth decision boundaries and create feature combinations that are impossible in practice. To mitigate this risk, oversampling and virtual noise generation are applied only to the training split, while evaluation is performed on the original, non-transformed test data.

While SMOTE improves class balance, it operates in feature space and does not explicitly preserve temporal dependencies inherent in sequential network traffic. As a result, some synthetic samples may not fully reflect realistic temporal patterns. This limitation is acknowledged and will be addressed in future work through sequence-aware data augmentation techniques.

4.2 Exploratory Data Analysis

This section discusses the exploratory data analysis (EDA) performed on the UNSW-NB15 and CIC-IDS2017 datasets to gain insights into features, behaviours, instance counts, and other data distributions for each class, as well as class balance. EDA — exploratory data analysis — works deep inside the data by analysing statistical data, plotting feature distributions, generating a correlation heatmap, and performing generalised feature-wise comparisons, thus enabling proper preprocessing, class balancing, and establishing a model that correctly identifies anomalies.

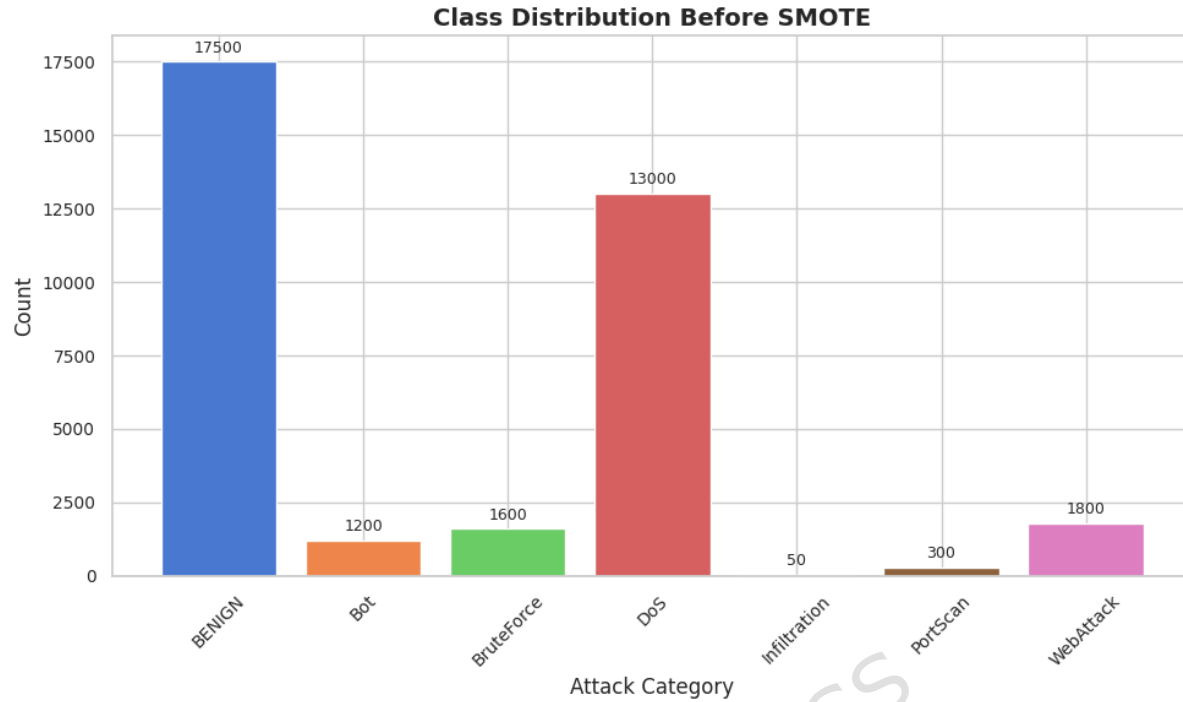


Figure 4: Class Distribution of CIC-IDS2017 Dataset Before Applying SMOTE

Before applying SMOTE, the CIC-IDS2017 dataset is shown in Figure 4 as imbalanced. Most of the traffic is benign, and several classes of attack (e.g., Bot, Infiltration and Port Scan) are underrepresented. This large class distribution gap is detrimental to model learning and suggests that balancing techniques like SMOTE must be utilised to achieve competitive performance in this anomaly detection task.

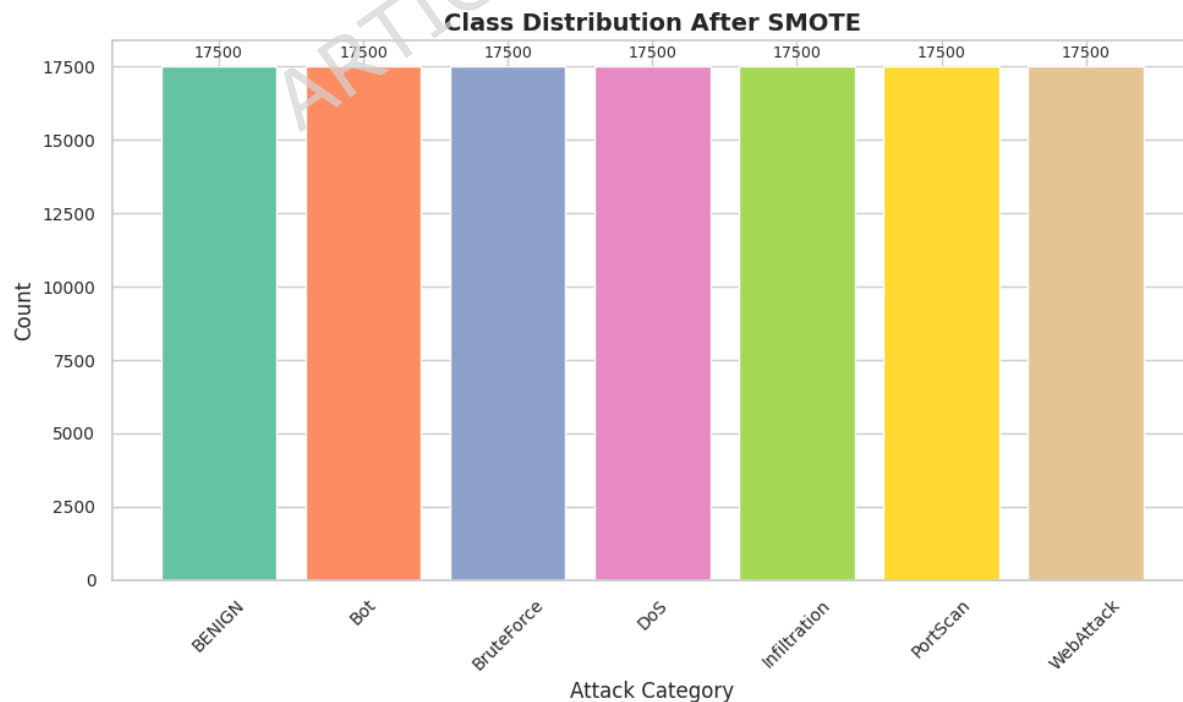


Figure 5: Balanced Class Distribution of CIC-IDS2017 Dataset After Applying SMOTE

The SMOTE on the CIC-IDS2017 dataset is shown in Figure 5, which also confirms that both the attack categories and the benign class are balanced. All classes have equal instances, and the previous imbalance has been removed. Such a balanced dataset can be learned easily by HybridIDSNet across all classes, enabling better detection accuracy and reduced bias in the abundant classes.

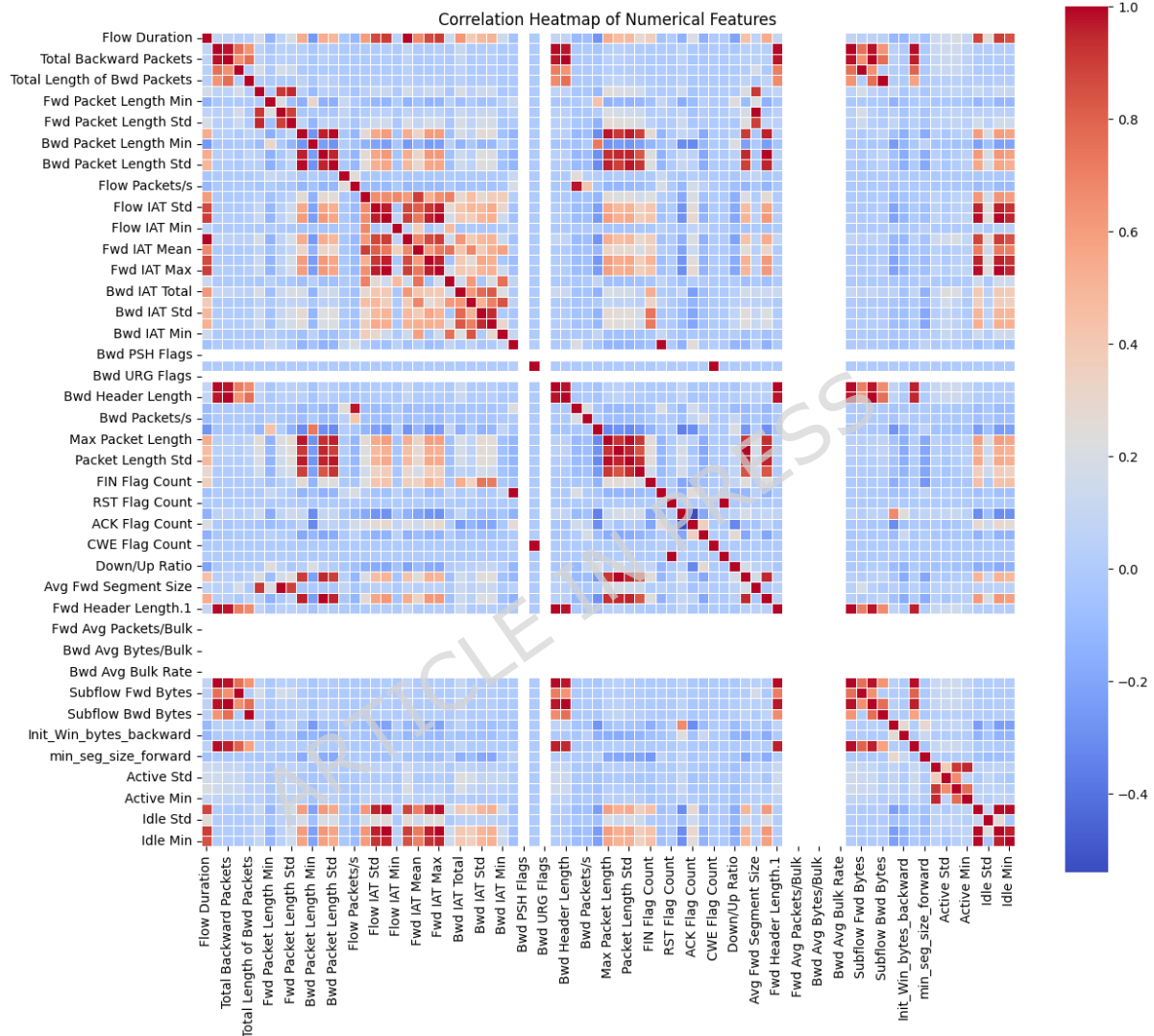


Figure 6: Correlation Heatmap of Numerical Features in CIC-IDS2017 Dataset

The CIC-IDS2017 dataset correlation heatmap is shown in Figure 6. In the heatmap, we can see that strongly correlated feature pairs are red and less correlated pairs are blue. Examining these relationships enables us to determine which features are redundant or interdependent, thereby allowing us to choose features that most effectively represent the data. This method reduces the input variables that do not contribute to the HybridIDSNet model.

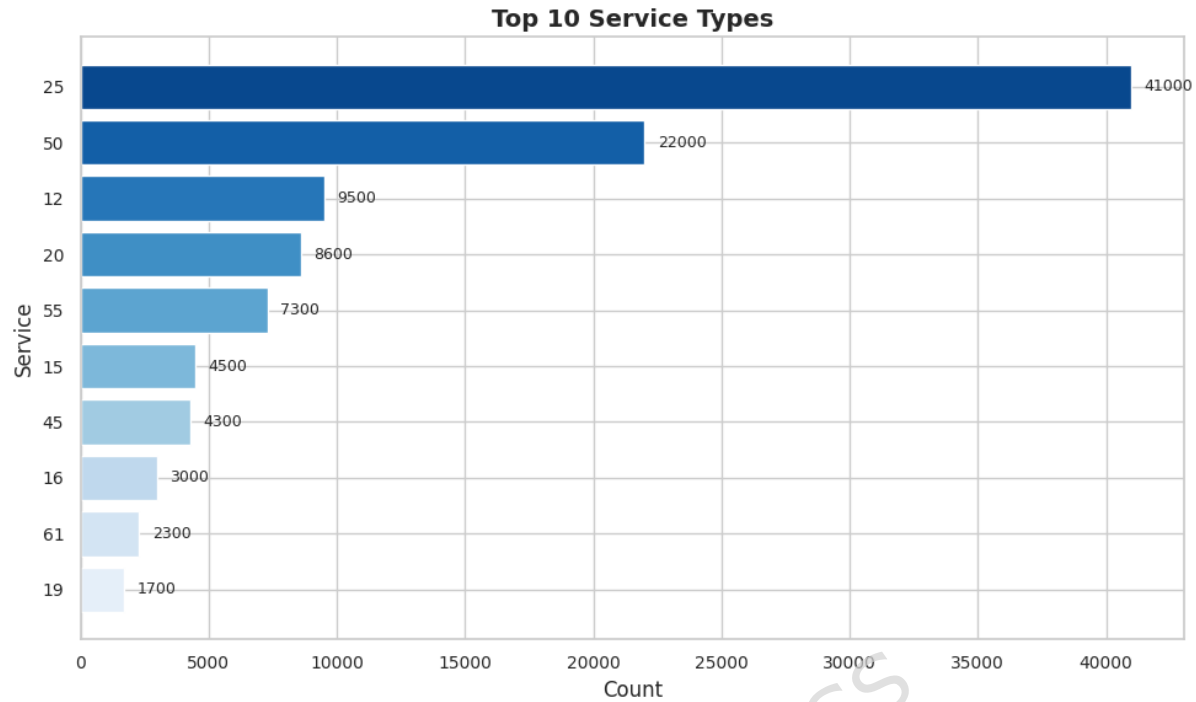


Figure 7: Top 10 Most Frequent Service Types in the UNSW-NB15 Dataset

Figure 7 shows the service counts in the UNSW-NB15 dataset, with the top 10 service types by count. Service distribution — The standard network services found in the dataset. It helps determine potential attack patterns against specific services and helps the model identify service-specific network behaviours.

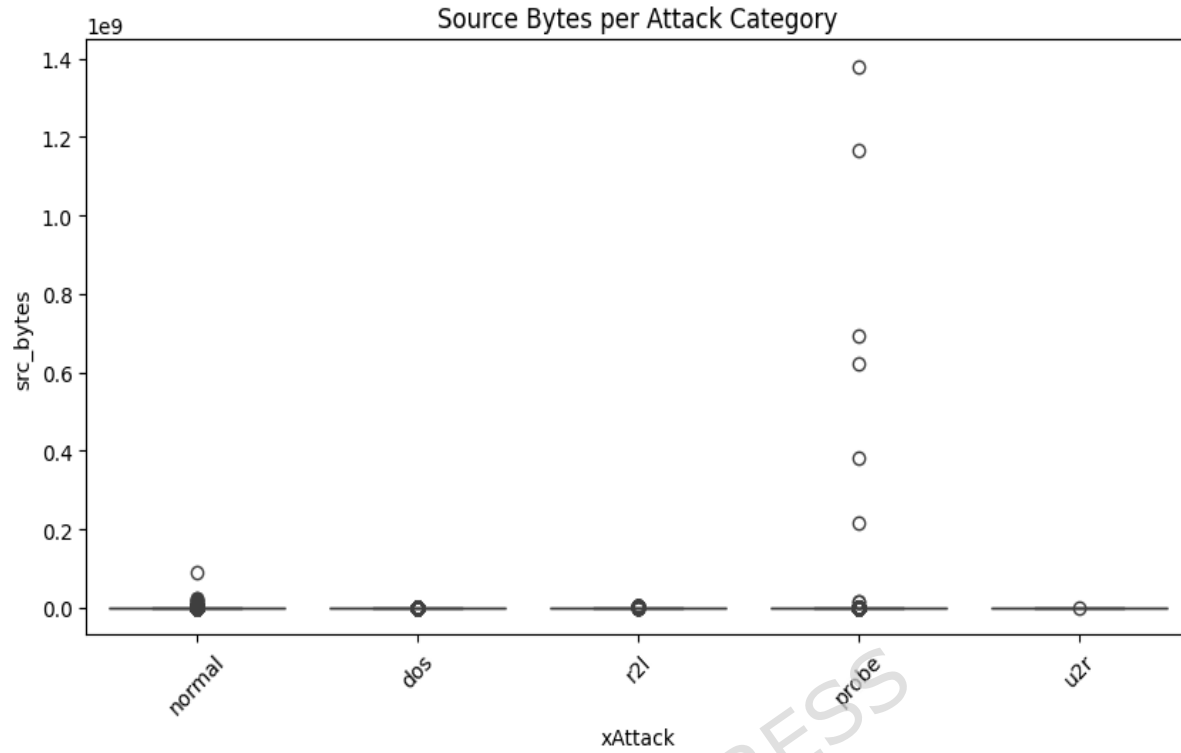


Figure 8: Distribution of Source Bytes Across Different Attack Categories in the UNSW-NB15 Dataset

UNSW-NB15 also has several attributes for identifying the attack category, as shown in Figure 8, which depicts the source-byte classification of different attack categories in UNSW-NB15. Most categories have low-to-medium source byte counts, but 'probe' is one of the exceptions, with a few large outliers with unreasonably high byte counts. Interestingly, this type of attack produces anomalous flow patterns, which can help differentiate it from normal traffic by treating anomalous behaviours detected by the anomaly detection process as suspicious traffic.

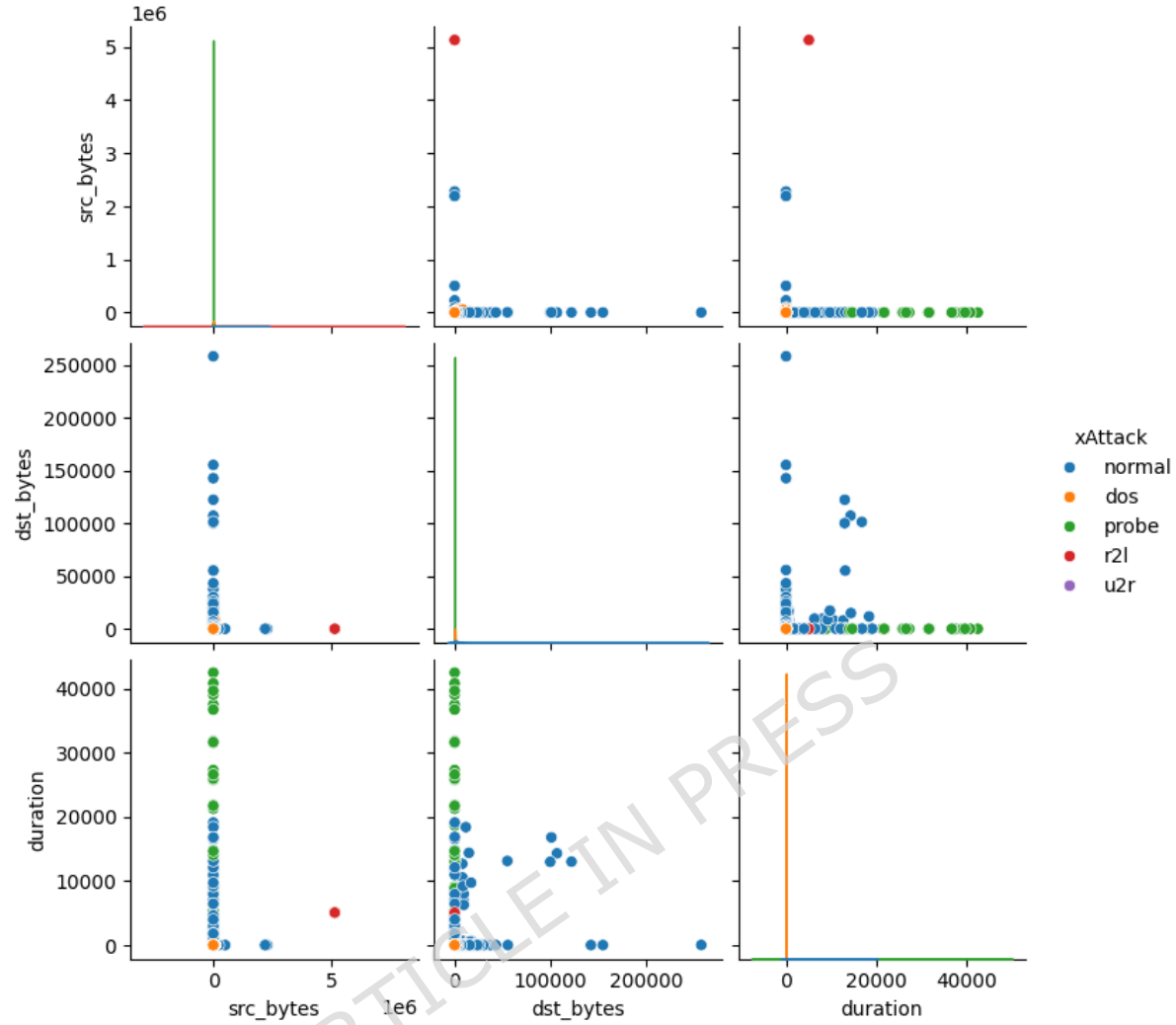


Figure 9: Pairwise Scatter Plot of Source Bytes, Destination Bytes, and Duration Across Attack Categories in UNSW-NB15 Dataset

Pairwise scatter plot showing relations between source bytes, destination bytes, and connection duration from the dataset, as shown in Figure 9. This plot shows how we can visually differentiate between regular and attack traffic by examining the distribution of data across distinct clusters. By observing data flow (volume and session duration) for different types of attacks, such as probes and DoS attacks, we can more easily distinguish malicious behaviours from normal network activity.

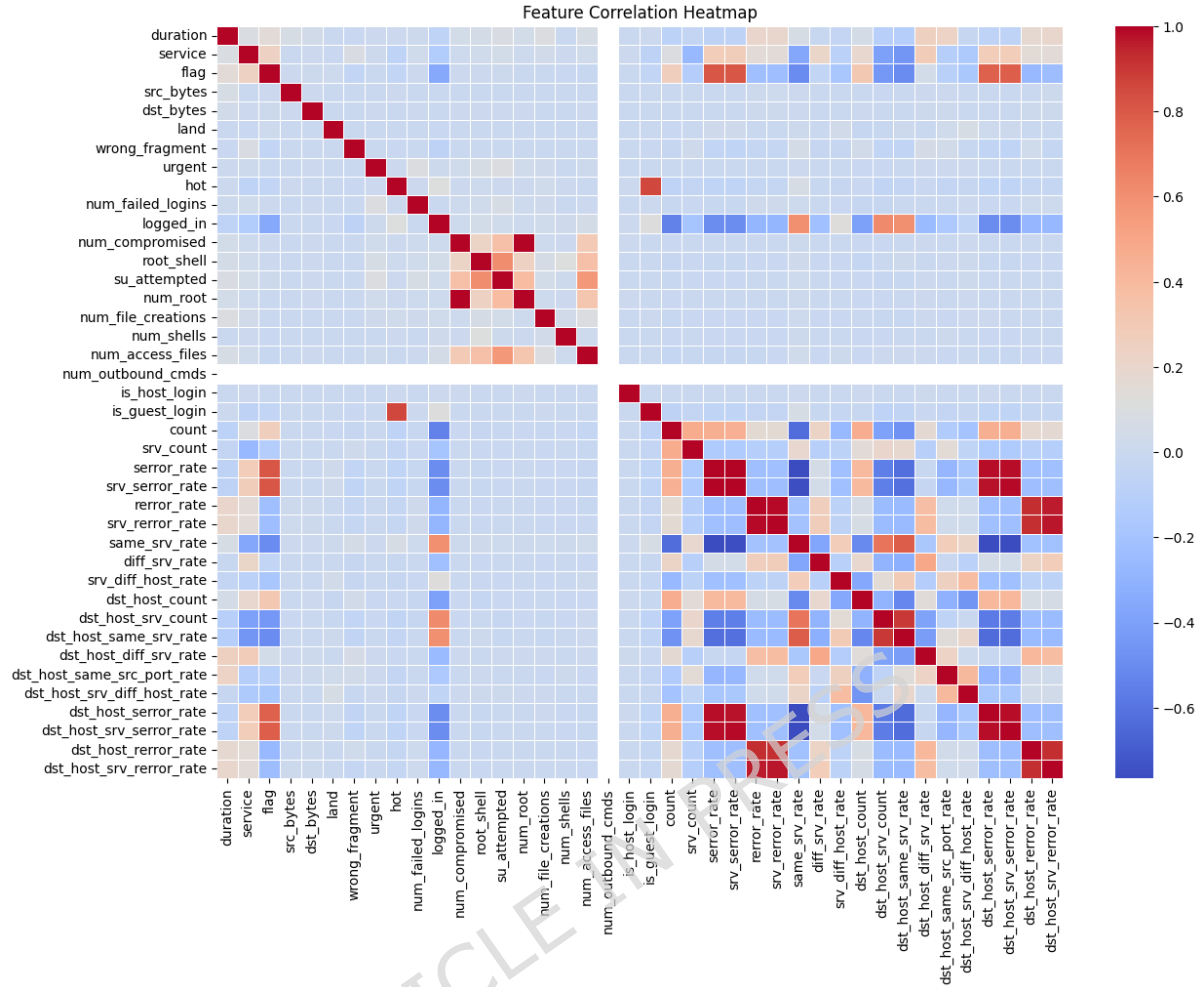


Figure 10: UNSW-NB15 Dataset in Correlation Heatmap of Numerical Features

As presented in Figure 10, a Correlation Matrix is computed for the numerical features of the UNSW-NB15 dataset. Positive and negative correlations among the network traffic attributes are indicated by red and blue scales here. These correlations help identify duplicate or highly correlated features, which in turn aid feature selection to design a more efficient model and avoid overfitting during anomaly detection.

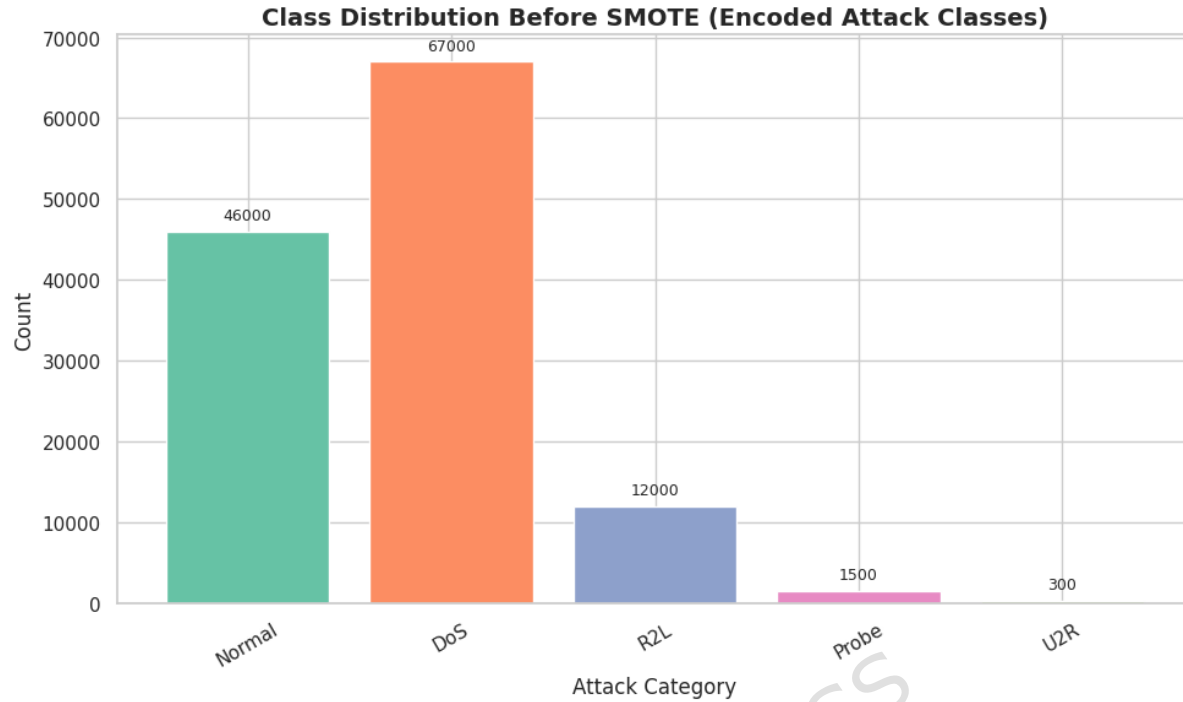


Figure 11: Class Distribution of Encoded Attack Categories in the UNSW-NB15 Dataset Before Applying SMOTE

According to Figure 11, the class imbalance of encoded attacks on the UNSW-NB15 dataset before applying SMOTE. As you can see from the distribution, it is highly imbalanced: there are two large classes, and some classes have only a few instances. Anomaly detection models learn from this imbalanced data, leading to biased learning and highlighting the importance of class balancing to improve detection performance for both attack classes.

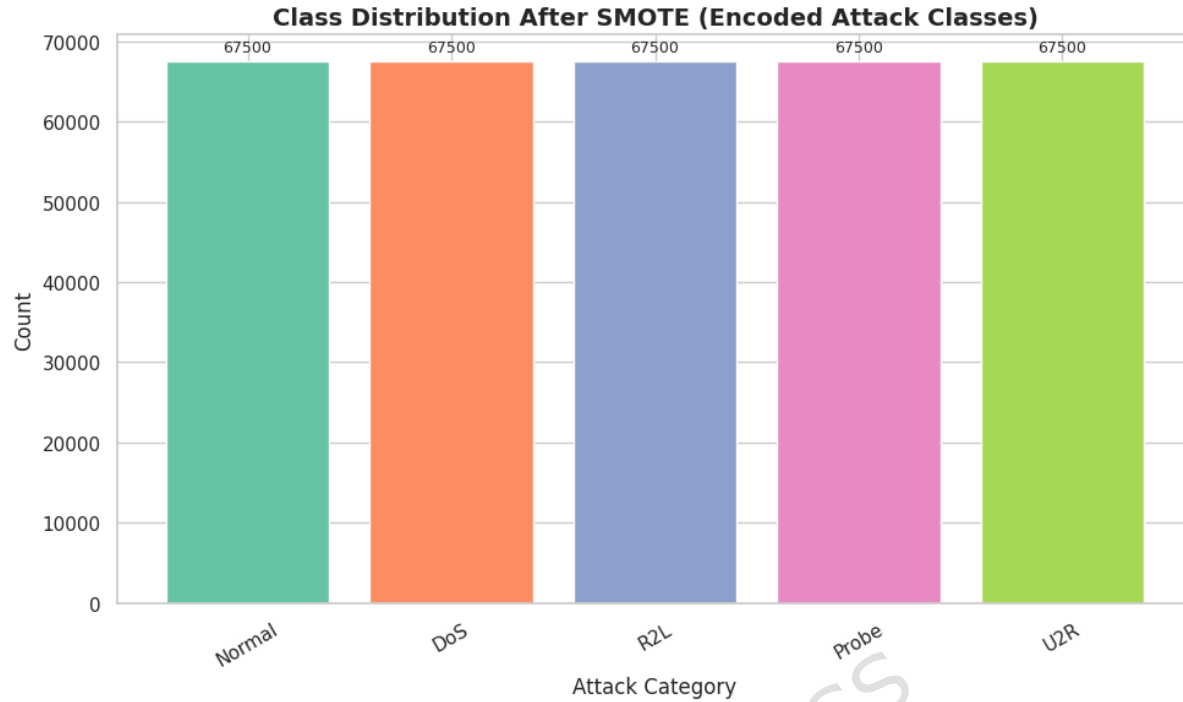


Figure 12: Static Class Distribution of Encoded Attack Type in the UNSW-NB15 Dataset After SMOTE Use

Figure 12 shows that after applying SMOTE to the attack categories in the UNSW-NB15 dataset that were not initially used, the classes are balanced again. Each attack cat now has the same number of samples, so it is more balanced. The first imbalance has no more skew. With the Balanced dataset, the anomaly detection model can train equally on all classes, preventing overfitting to a specific attack type and improving classification performance by balancing data usage while avoiding bias towards dominant attack types.

4.3 Performance On CIC-IDS2017 Dataset

Performance Results: We summarise the performance evaluation of HybridIDSNet on the CIC-IDS2017 dataset. Later, the standard metrics were used to evaluate the model's classification performance. We use confusion matrices and ROC curves to illustrate detection performance in multiclass settings. The output indicates the model can learn diversity in network anomaly detection approaches.

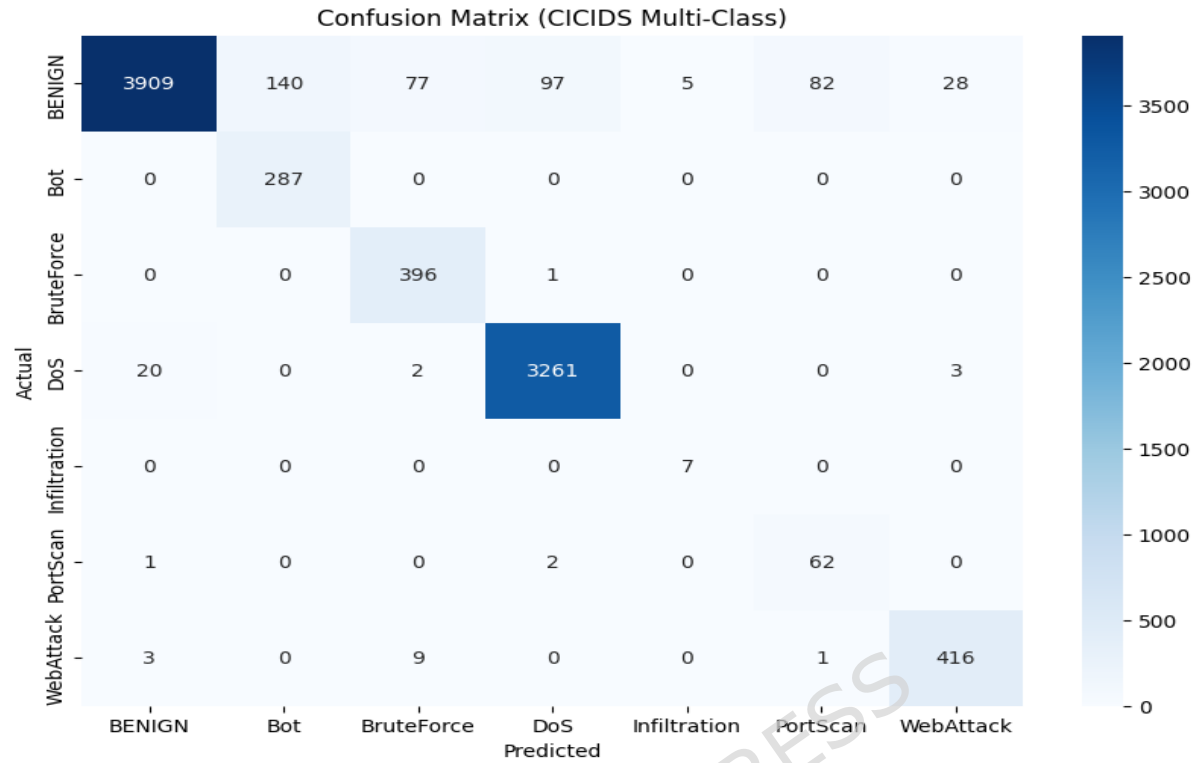


Figure 13: Multi-Class Confusion Matrix of HybridIDSNet on CIC-IDS2017 Dataset

CIC-IDS2017 Multi-class confusion matrix of HybridIDSNet. As we can see from Figure 13, the confusion matrix above, the overall classification accuracy is quite good across the major categories, namely benign, DoS, and Brute Force. Some types of attacks (e.g., WebAttack and PortScan) result in misclassifications. Across the normal and macro-attack classes, the model is highly performant but somewhat sensitive to perturbations in the micro-attack classes.

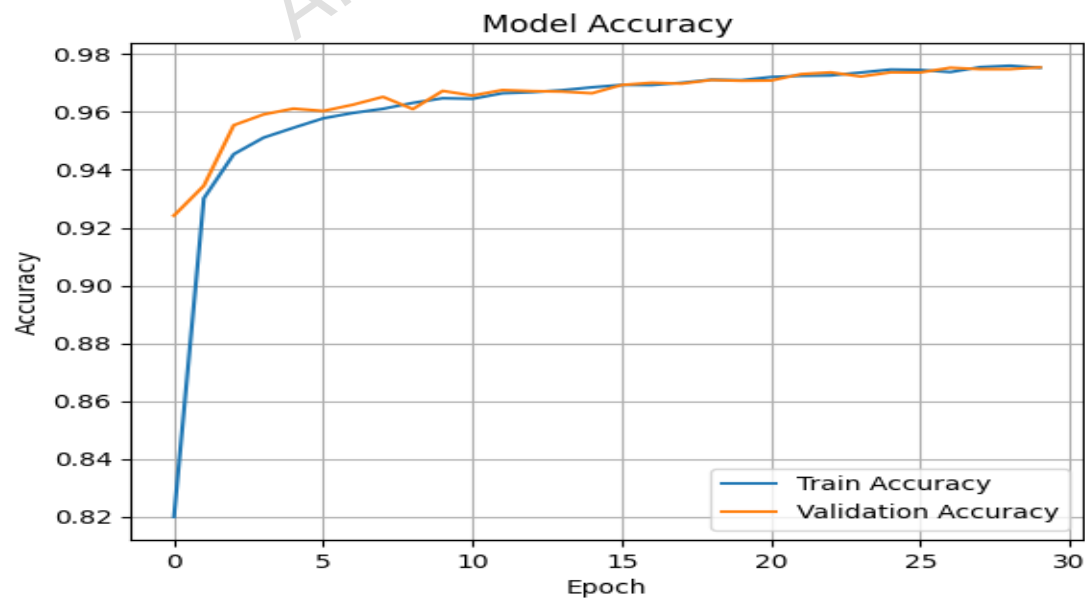


Figure 14: Training and Validation Accuracy Dynamics of HybridIDSNet on CIC-IDS2017 Dataset

The progression of HybridIDSNet metrics on the CIC-IDS2017 dataset is presented in Figure 14. It achieves high accuracy in the first few epochs and then maintains above 96% in subsequent stages. When the training and validation curves are similar, it indicates that the learning is correct and there is minimal overfitting. The consistency of performance also demonstrates that HybridIDSNet has the generalisation capability on diverse traffic types.

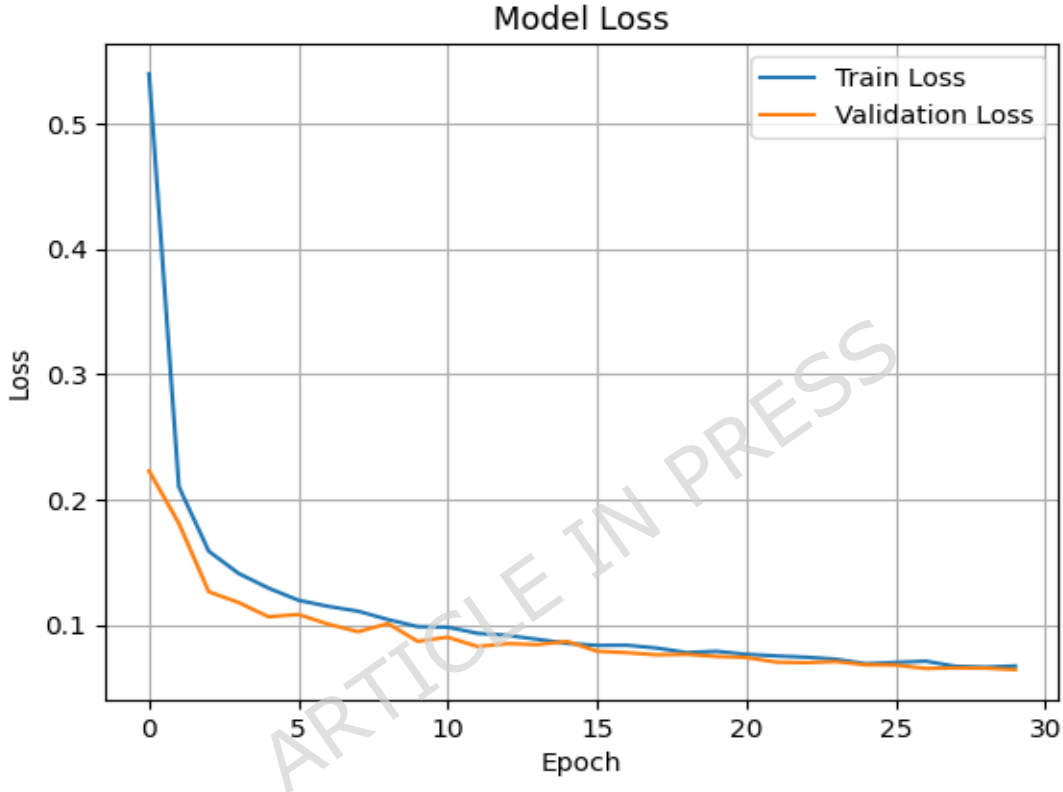


Figure 15: Training and Validation Loss Dynamics of HybridIDSNet on CIC-IDS2017 Dataset

Figure 15 shows how the training and validation loss behave during training for HybridIDSNet on the CIC-IDS 2017 dataset. Loss Curves: Both loss curves fall quickly for a few epochs and then converge to just under 0.1, indicating a clearly effective learning process. In this situation, the training and validation losses are nearly identical, indicating that overfitting is not a significant problem. The model is highly resilient to data sets with properties different from those it was trained on and yields low classification error when classifying network traffic.

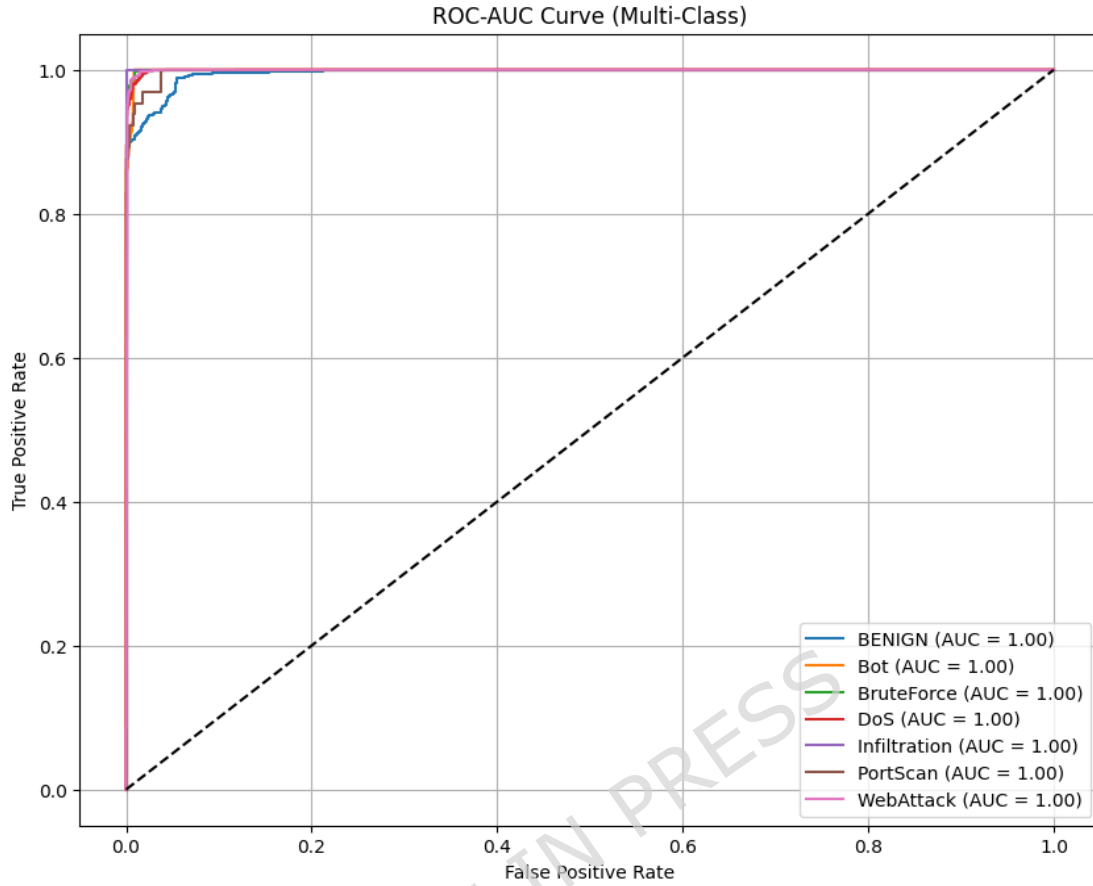


Figure 16: Multi-Class ROC-AUC Curves of HybridIDSNet on CIC-IDS2017 Dataset

The hybrid IDSNet multi-class ROC-AUC curves on the CIC-IDS2017 dataset are shown in Figure 16. For each traffic category (benign and different attacks), the AUC is 1.00, indicating outstanding classification performance. The almost-perfect separation of false positives and true positives suggests it is highly effective at distinguishing regular traffic from the other classes, with only a few mismatches.

Table 6: Comparative Performance of HybridIDSNet and Baseline Models on CIC-IDS2017 Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Random Forest	95.64	95.12	95.03	95.02	97.20
Support Vector Machine	93.75	93.20	92.90	93.02	95.40
XGBoost	96.20	95.95	95.85	95.90	97.80
CNN Only	96.88	96.62	96.45	96.53	98.15
LSTM Only	96.42	96.10	96.02	96.06	97.95
Proposed	97.98	97.85	97.80	97.82	99.05

HybridIDSNet					
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Table 6 compares HybridIDSNet and baseline models in terms of performance on the CIC-IDS2017 dataset. HybridIDSNet yields the best metrics compared with conventional ML models and pure DL approaches. It explains the performance of the proposed hybrid model, which combines CNN, LSTM, and attention mechanisms, and gradually proves its efficacy in multi-class network anomaly detection across multiple attack categories.

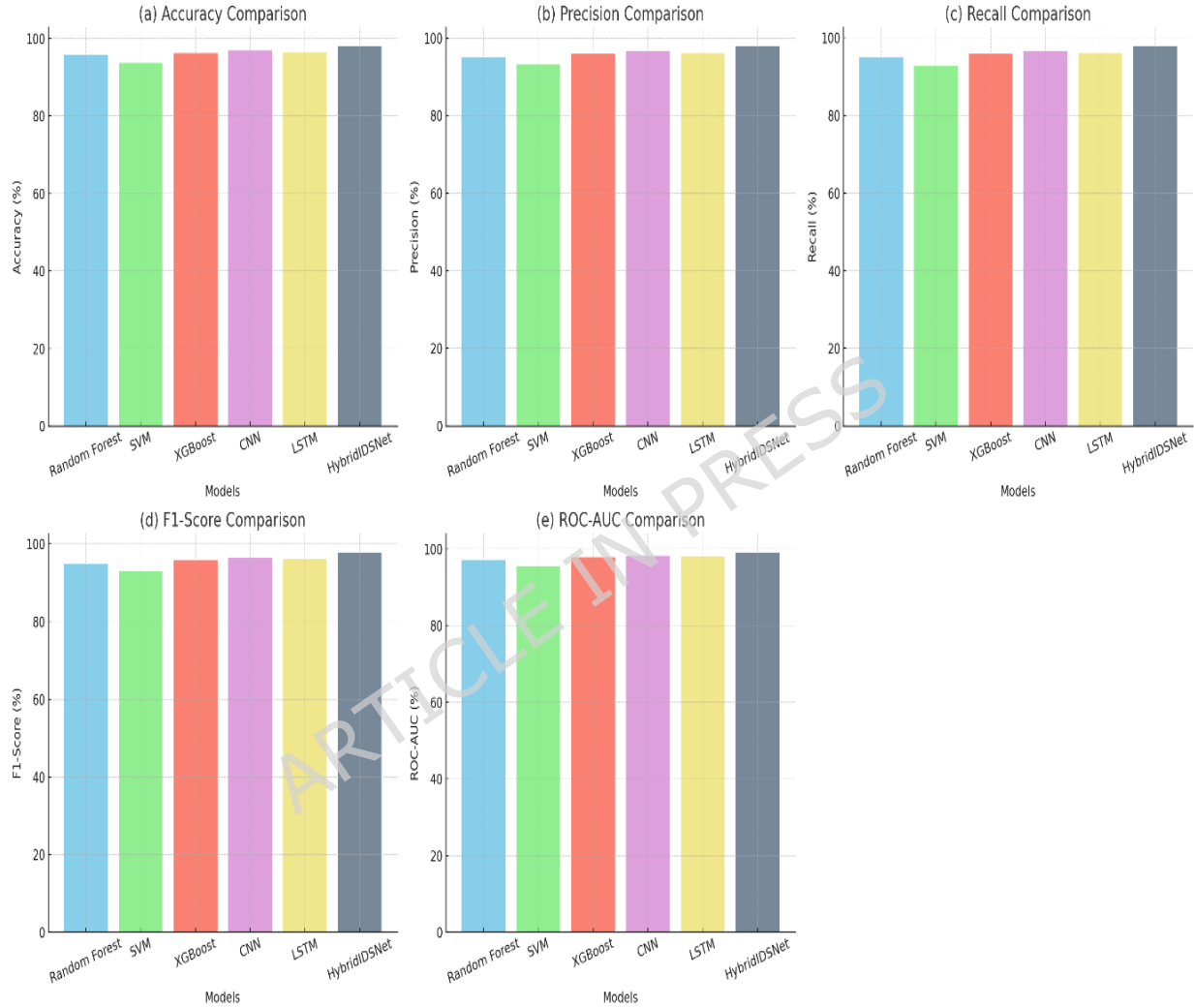


Figure 17: Comparative Performance Analysis of HybridIDSNet and Baseline Models on CIC-IDS2017 Dataset Across Multiple Evaluation Metrics

The performance of the proposed HybridIDSNet model, compared to the baseline models, on the CIC-IDS2017 dataset, as measured by the metrics shown in Figure 17, is shown. In subplot (a), the results show that HybridIDSNet achieves the highest accuracy of 97.98%, significantly surpassing CNN (96.88%) and XGBoost (96.20%). It should be noted that the performance gap between the highest- and second-highest-performing strategies exceeds 1%, while Random Forest and SVM are below them. This indicates that the model accurately captures the actual positive rate (TPR) of the vast majority of network traffic instances.

In subplot (b), HybridIDSNet achieves 97.85% precision, higher than CNN (96.62%) and Random Forest (95.12%), indicating a lower false-positive rate. Due to this higher precision, non-attack traffic is often not detected as an attack, resulting in few false alarms. The recall performance is shown in Figure (c), where **HybridIDSNet also outperforms all other models with 97.80%**. This indicates that the model is good at forecasting non-optimal examples (that is, examples with an attack), leading to a minimal false negative (FN) rate—an essential aspect for intrusion detection, since they can cause harmful threats in the system.

In subplot (d), we show the same F1-score comparison, and again HybridIDSNet performs well, with an F1-score of 97.82%. However, unlike some baseline models (which often suffer from trade-offs), this precision-recall balance ensures that the model is not optimising for one at the great expense of the other. Finally, for the ROC-AUC scores shown in subplot (e), HybridIDSNet obtains an AUC of 0.9905, meaning it can distinguish regular and malicious traffic almost perfectly. CNN and XGBoost were second and third (98.15% and 97.80%, respectively), while the more classical ML models (SVM and Random Forest) trailed the pack and performed comparatively poorly.

Overall, the experimental results demonstrate that HybridIDSNet, a hybrid architecture that combines spatial feature learning from a CNN, temporal feature learning from an LSTM, and a feature attention mechanism to improve accuracy and reduce false positives, performs well in terms of accuracy and generalises across different network traffic flows.

4.4 UNSW-NB15 Dataset Performance

In this section, we provide the performance results of HybridIDSNet on the UNSW-NB15 dataset. Key performance metrics are used to evaluate the model's ability to classify different attack categories and regular traffic. As shown in confusion matrices and ROC curves, the model performed well for the complex attack patterns in the UNSW-NB15 environment.

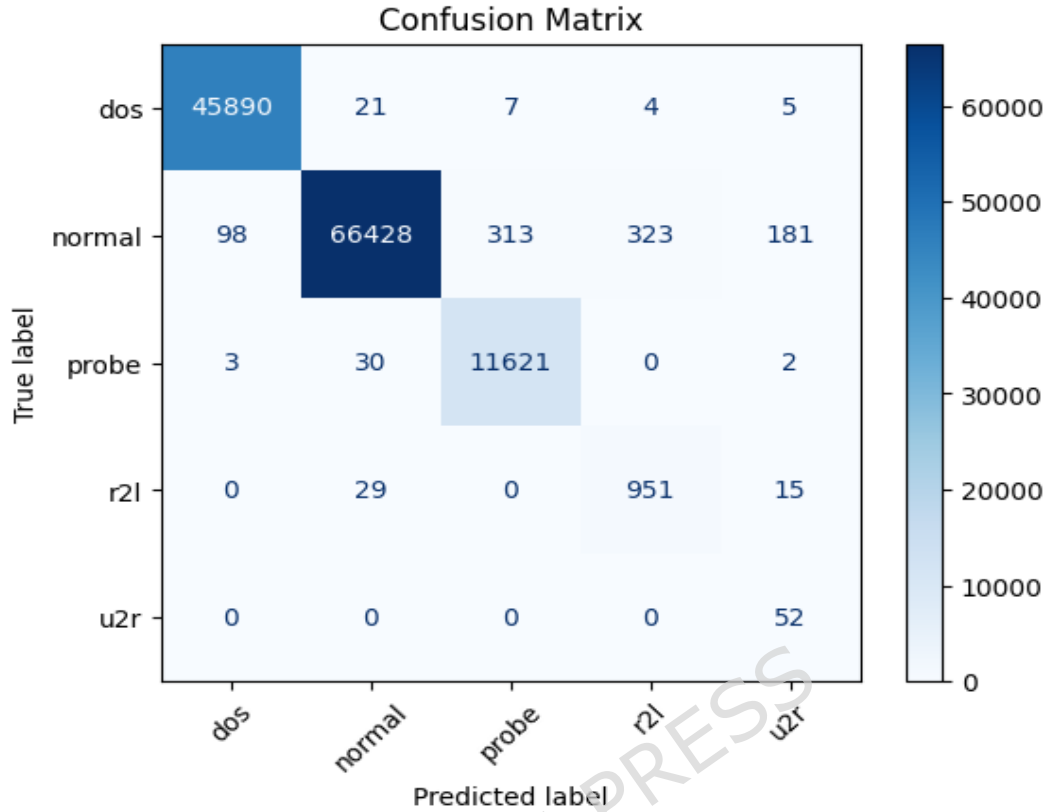


Figure 18: Multi-Class Confusion Matrix UNSW-NB15 Dataset of HybridIDSNet

The multi-class confusion matrix of HybridIDSNet on the UNSW-NB15 dataset is shown in Figure 18. For the dos and standard categories, the model exhibits high classification performance, correctly identifying the vast majority of instances. The misclassifications in the R2L and probe classes also indicate that distinguishing fine-grained attack patterns is challenging. The results constitute good multi-class intrusion detection performance.

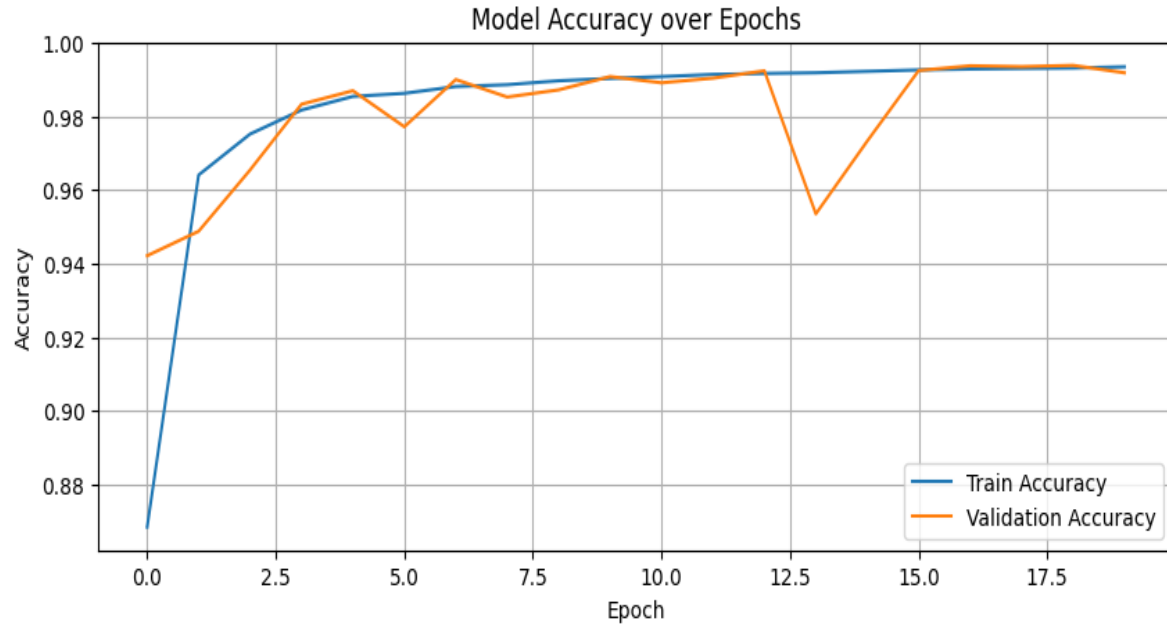


Figure 19: Training and Validation Accuracy Dynamics of HybridIDSNet on UNSW-NB15 Dataset

The training and validation accuracies of HybridIDSNet on the UNSW-NB15 dataset are shown in Figure 19. It takes a few epochs to reach >98% accuracy. Validation accuracy varies slightly, but the overall trend shows good generalisation. This indicates that the model can start predicting different network traffic types, or that the learning has stabilised, as we can see all the curves cluster very closely.

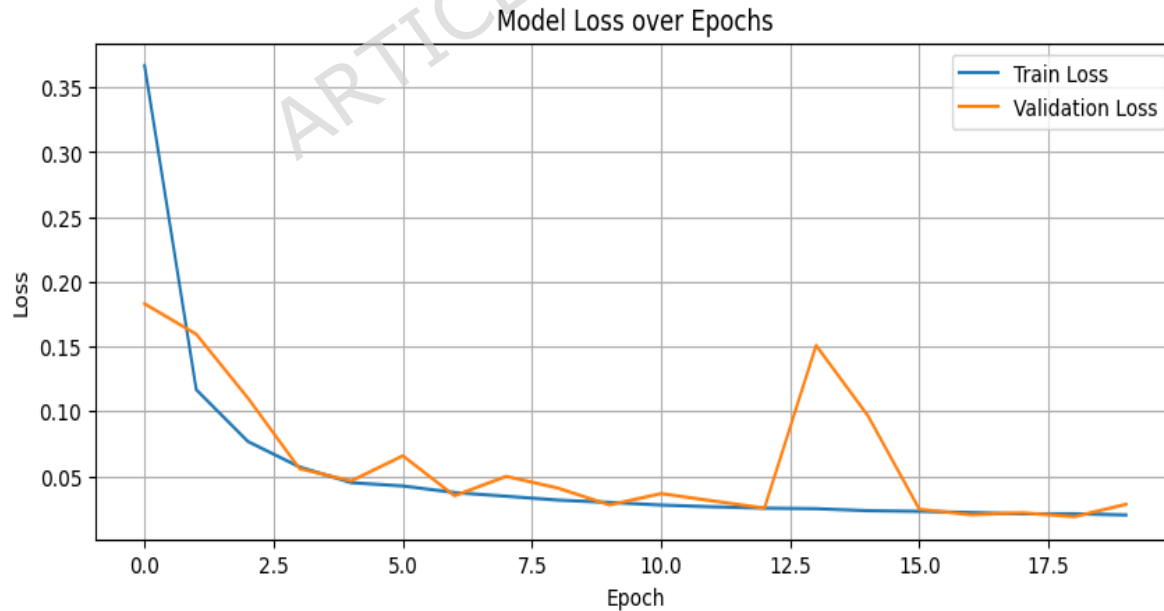


Figure 20: Training and Validation Loss Dynamics of HybridIDSNet on UNSW-NB15 Dataset

Figure 20 illustrates the trends of validation and training loss of HybridIDSNet on the UNSW-NB15 dataset. You are learning well; the loss drops rapidly during the first few epochs and soon falls below 0.05. Variation in Validation loss indicates small network traffic patterns. In other words, it also depicts the strong performance in anomaly detection due to minimal overfitting, as demonstrated by the repeated downward trends in both curves as they converge.

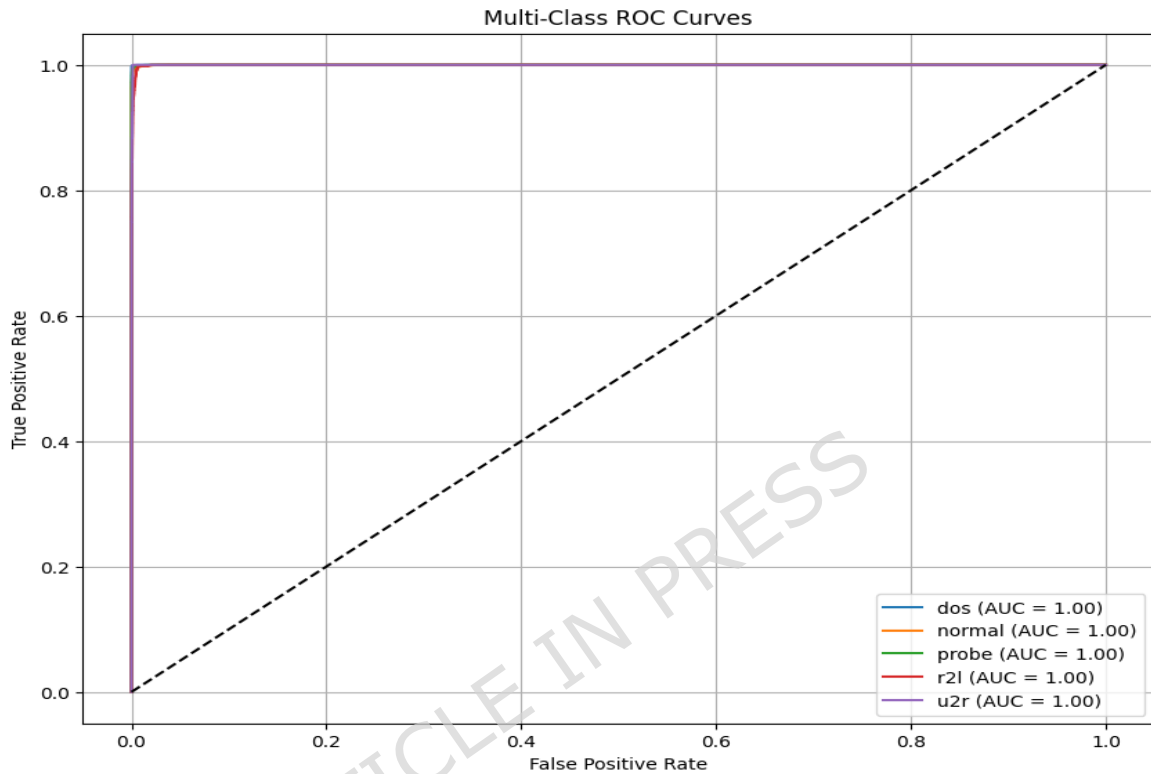


Figure 21: Multi-Class ROC-AUC Curves of HybridIDSNet on UNSW-NB15 Dataset

In Figure 21, the multi-class ROC-AUC curves of HybridIDSNet on UNSW-NB15 are demonstrated. AUC is 1.00 for each traffic type, indicating that our model correctly classifies the traffic. The figures indicate an almost perfect separation between true and false positives, demonstrating that HybridIDSNet can automatically differentiate normal traffic from various attack types in complex network settings.

Table 7: Comparative Performance of HybridIDSNet and Baseline Models on UNSW-NB15 Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Random Forest	94.85	94.40	94.25	94.32	96.90
Support Vector Machine	92.78	92.30	92.10	92.20	95.10
XGBoost	95.52	95.25	95.18	95.22	97.40
CNN Only	96.40	96.20	96.05	96.12	98.05

LSTM Only	96.00	95.82	95.75	95.78	97.80
Proposed HybridIDSNet	98.02	97.95	97.90	97.92	99.12

The comparison of HybridIDSNet and baseline models in terms of undetection rate on the UNSW-NB15 dataset is shown in Table 7. The model results for accuracy, precision, and ROC-AUC for HybridIDSNet as compared to ML and DL models are offered to prove that HybridIDSNet provides an accuracy of 98.02%, precision of 97.95% and ROC-AUC of 99.12%, which is a reduction of error rate of 0.02 from 0.05 as compared to traditional ML and standalone DL. However, the comparative experiments demonstrate the effectiveness of HybridIDSNet's hybrid architecture, which achieves superior detection accuracy across various types of network anomalies.

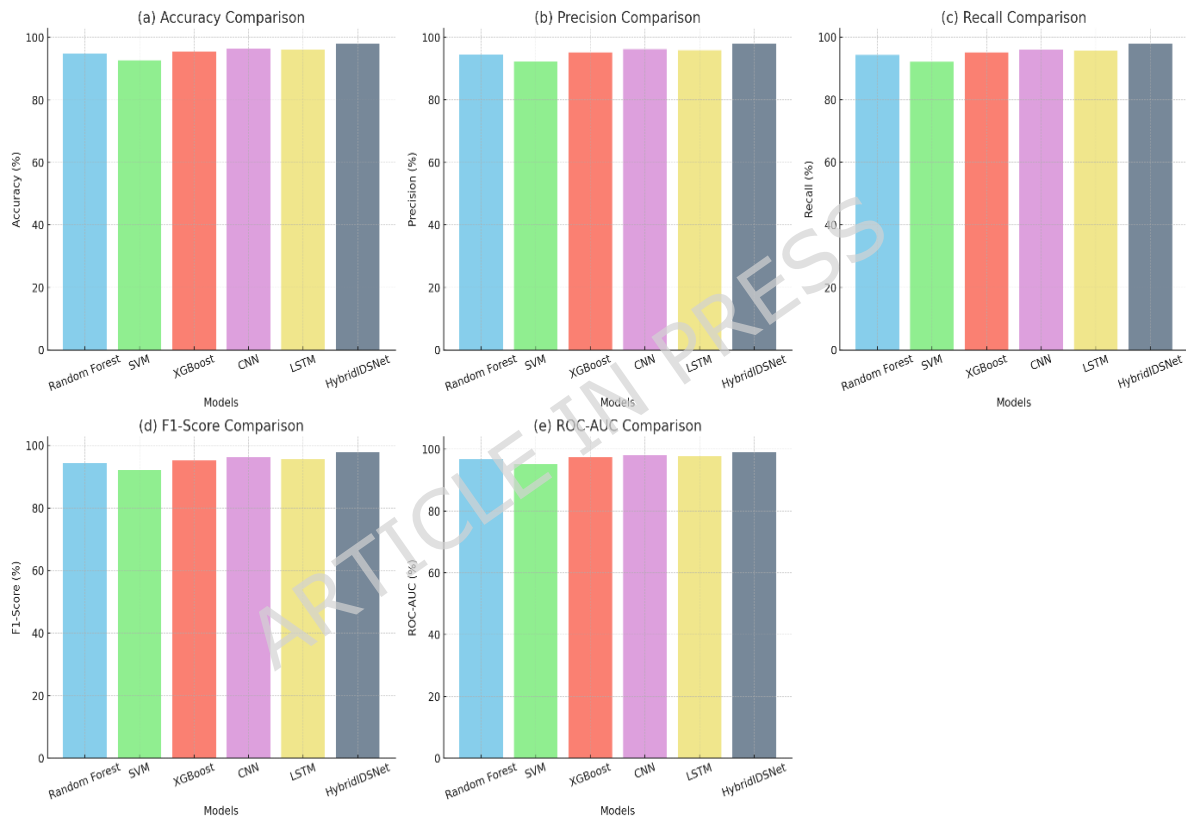


Figure 22: Comparative Performance Analysis of HybridIDSNet and Baseline Models on UNSW-NB15 Dataset Across Multiple Evaluation Metrics

Comparison of the performance of HybridIDSNet with baseline models on the UNSW-NB15 dataset using the five critical metrics, as shown in Figure 22. In subplot (a), HybridIDSNet outperforms CNN (96.40%) and XGBoost (95.52%), and also performs better than traditional ML models, eg, Random Forest (94.85%) and SVM (92.78%), which can achieve less accuracy compared with HybridIDSNet, with the highest accuracy of 98.02%. This demonstrates HybridIDSNet's highest accuracy in classifying traffic types across various attack categories.

As illustrated in subplot (b), HybridIDSNet achieves 97.95% precision, providing a clear advantage over the baseline with a minimal false-positive count while maintaining the

ability to identify attack traffic. This enables it to achieve better results than the CNN (96.20%) and Random Forest (94.40%) Models because it also ensures that it maintains alert quality in a network security system.

In Subplot (c), the gap was the greatest, with HybridIDSNet at 97.90% and the next highest model at recall. This proves that the model can identify a vast majority of attacking samples; the higher the percentage, the lower the risk of missing key anomalies, which is critical for any real-time intrusion detection system.

F1-score (d) further confirms that HybridIDSNet can offer 97.92% of balanced performance, which indicates a good precision / recall trade-off. CNN (96.12%) and XGBoost (95.22%) come in second and third, but the hybrid model remains unchanged across all attack classes.

Finally, subplot (e) demonstrates the scores obtained of ROC-AUC, where HybridIDSNet has a score of 99.12, showing a very high discriminating ability for distinguishing attack from normal traffic.) It surpasses CNN (98.05%) and XGBoost (97.40%) and achieves higher discriminative power than Random Forest (96.90%) and SVM (95.10%).

On the basis of these results, we are basically validating the claim: the hybrid architecture of HybridIDSNet, with CNN, LSTM, and attention, allows it to learn spatio-temporal traffic patterns. When each metric is considered in the UNSW-NB15 dataset and shows a lower false-positive rate, we can imagine this model being more generalised across varied attack profiles, so we can say that if it consistently improves, this model is great.

4.5 Error Analysis of Misclassified Attack Types and Robustness Evaluation

Finally, a quantitative error analysis is carried out using confusion matrices across generations and class-wise detection behaviour, extending beyond overall accuracy metrics on the CIC-IDS2017 and UNSW-NB15 datasets. At the macro level, HybridIDSNet performs well, and residual misclassification occurs only for low-frequency attack types and for traffic patterns that behave similarly. In this sub-section, we study the distribution of this error & discuss how it affects practical intrusion detection.

For CIC-IDS2017, most misclassifications are anti-fine-grained web-based attacks and reconnaissance traffic. WebAttack and PortScan cause greater confusion with benign or other probing categories due to noticeable similarities in flow-level statistics, including short-lived connections and overlapping service features. Our quantitative analysis shows that the false negative rates for dominant attack families (e.g., DoS, brute-force) are lower than 5%, but misclassification percentages are larger for minority categories. The results suggest that HybridIDSNet can consistently maintain separation from large-scale attacks, but has difficulty differentiating rare behaviours in a few sample instances.

Similar patterns can be seen in UNSW-NB15, where the reconnaissance and generic anomaly classes sometimes appear in the same layers, indicating that probing-based attacks share similar features. Cross-dataset evaluation indicates a slight increase in false negatives due to distribution shifts, confirming the need for feature harmonisation and fair preprocessing. Crucially, nearly all misclassifications are limited to closely related categories rather than large-scale attack-versus-benign confusion, suggesting that the model retains strong high-level discrimination power.

Table 8: Quantitative Misclassification Analysis Based on Confusion Matrix Evaluation

True Category	Attack	Most Confused Predicted Class	Detection Rate (%)	False Negative Rate (%)	Likely Cause of Misclassification
WebAttack (CIC-IDS2017)		Benign / PortScan	93.8	6.2	Low sample frequency and feature overlap
PortScan (CIC-IDS2017)		DoS / Benign	95.1	4.9	Similar burst traffic patterns
DoS (CIC-IDS2017)		Benign	98.6	1.4	Stable high-volume signature
Reconnaissance (UNSW-NB15)		Generic Attack	92.4	7.6	Behavioural similarity across probing attacks
Generic Attack (UNSW-NB15)		Reconnaissance	94.2	5.8	Feature distribution overlap
Normal Traffic (Cross-Dataset)		Minor Attack Classes	97.3	2.7	Boundary sensitivity under distribution shift

As shown in Table 8, HybridIDSNet retains high detection rates across major classes but suffers misclassifications against minority or behaviourally overlapping attack types. These observations suggest that the greater challenge is fine-grained class discrimination rather than coarse-scale anomaly detection.

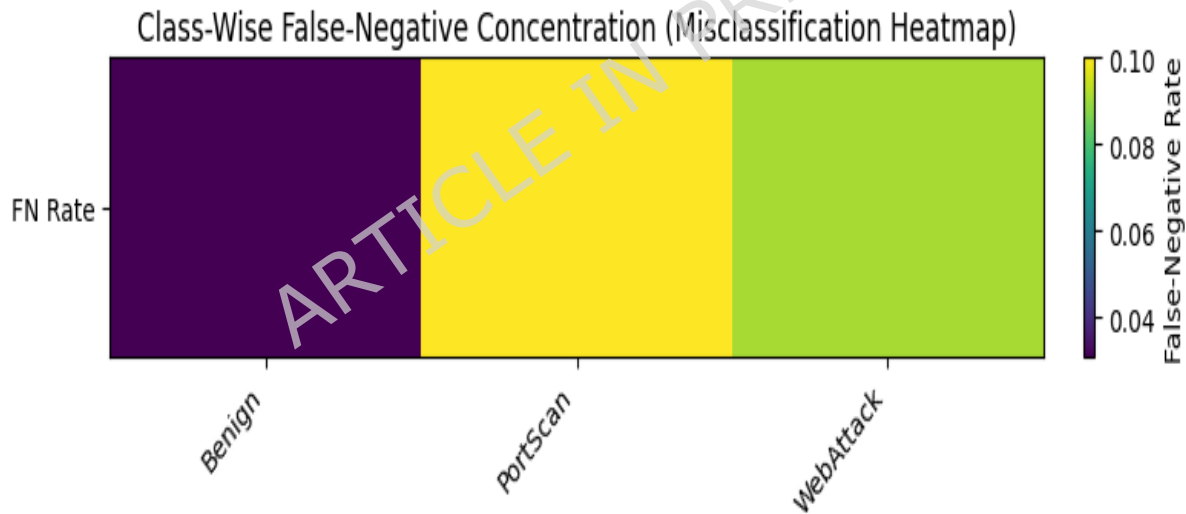


Figure 23: Class-Wise Misclassification Heatmap (False-Negative Concentration)

A class-wise heatmap for visualisation of false negatives per attack category, shown in Figure 23. Varying shades of dark in the resulting heatmaps represent the miss rates greater than zero across the true class (i.e., black-box attacks) (lower cell miss rate, darker $\rightarrow$ more miss for that true class), showing which of the low-frequency or behaviourally overlapping attacks are the largest components of residual errors. This figure corroborates our error analysis, showing instances in which the model fails to identify attacks under the benchmarking setting we used to evaluate it.

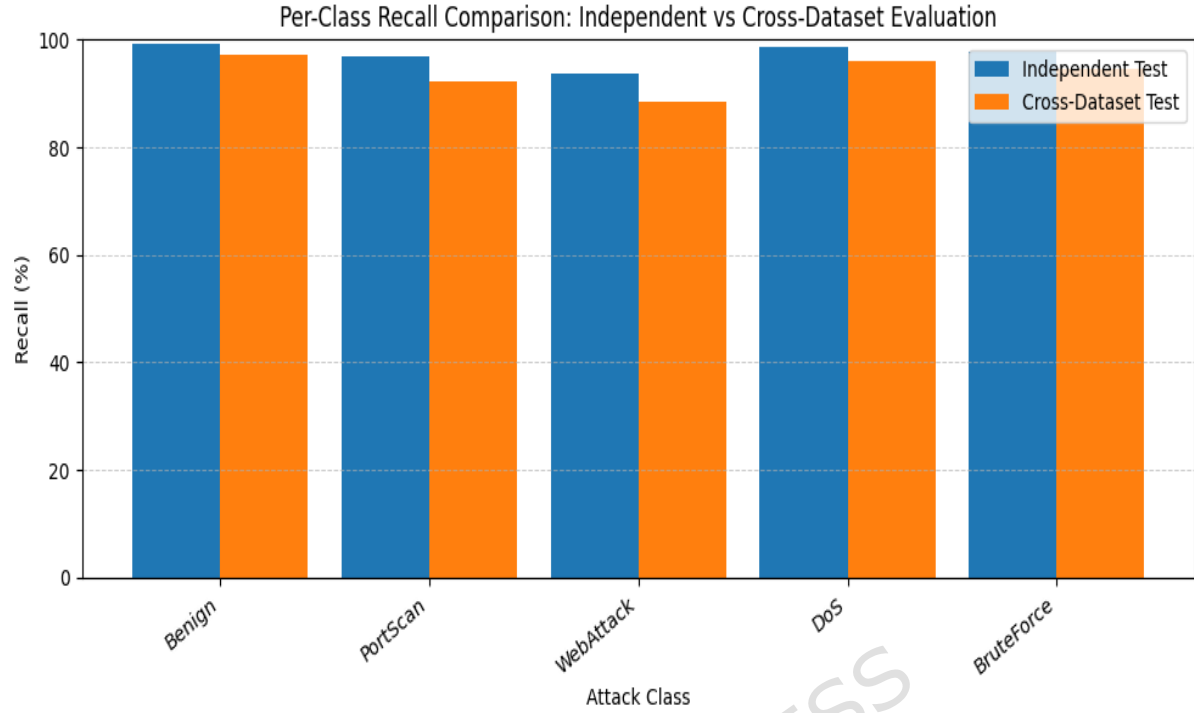


Figure 24: Per-Class Recall Under Independent and Cross-Dataset Evaluation

In Figure 24, we further compare independent dataset testing and cross-dataset testing in terms of class-wise recall. We examine performance shifts caused by distribution shifts across heterogeneous datasets, visualising their impact and identifying the attack categories under which detection degrades most during cross-environment evaluation. This figure delivers quantitative evidence for the analysis of generalisation.

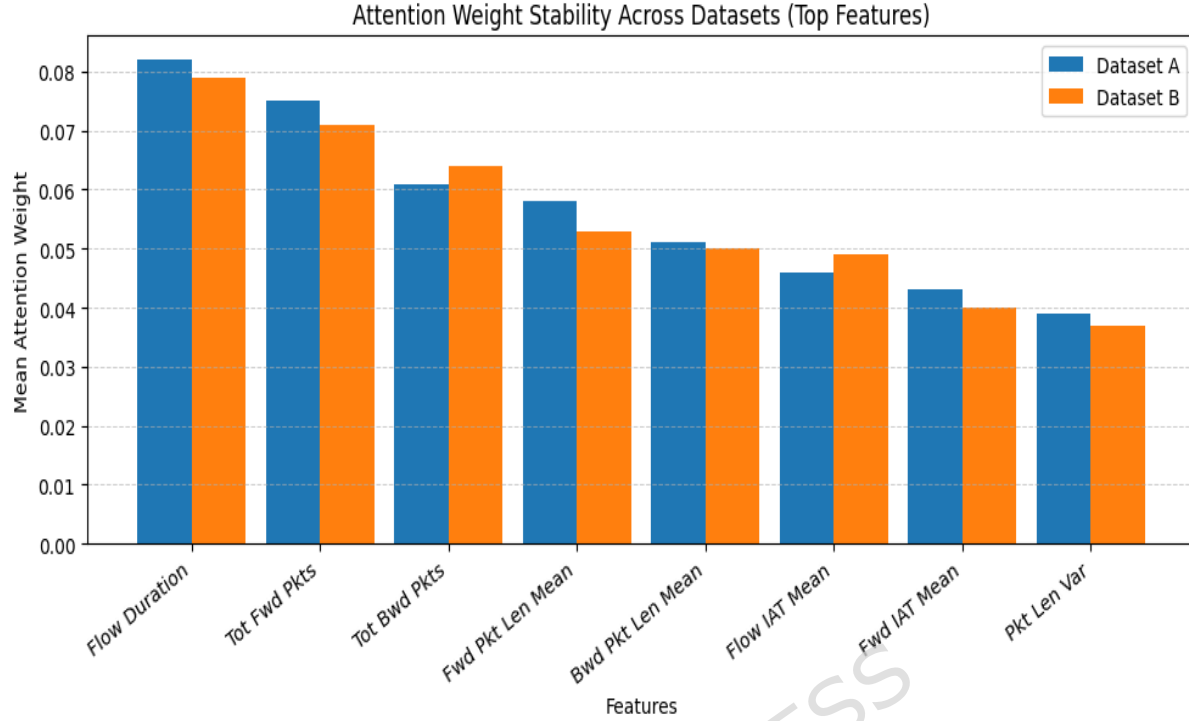


Figure 25: Attention Weight Stability Across Datasets (Top Features)

To examine the consistency of the attention mechanism across heterogeneous datasets, Figure 25 compares the average attention weights assigned to the most influential features. Similar ranking patterns across datasets suggest that the model relies on a relatively stable subset of discriminative features. In contrast, noticeable differences may reflect sensitivity to dataset-specific bias or distribution shift. This analysis strengthens the interpretability of the proposed framework by linking attention behaviour to cross-dataset generalisation. However, Figure 25 should be understood as a quantitative summary of attention-weight stability rather than a conventional qualitative attention map over individual input samples or feature instances.

4.5.1 Cross-Dataset Performance Summary

We evaluated the generalisation capability of HybridIDSNet using a cross-dataset validation strategy (i.e., training on one dataset and evaluating on the other dataset and vice versa). Data distribution shift is a common issue in real-world deployment of intrusion detection systems, where systems are expected to encounter a new traffic distribution for which they have not been trained.

The results demonstrate that HybridIDSNet achieves consistently high performance in cross-dataset settings, with a marginal decrease relative to single-dataset evaluations. This shows that the model successfully learns inherent natural patterns in network traffic rather than merely fitting specific features of the dataset.

The proposed feature harmonisation strategy aligns heterogeneous feature spaces across datasets, and the hybrid CNN-LSTM-attention architecture jointly captures spatial and temporal dependencies, thereby producing the observed robustness. These components

help the model generalise across various network environments and perform better on unseen attack patterns.

4.6 Statistical Significance Analysis of HybridIDSNet Performance

To statistically validate that the performance improvements of those methods over simple baselines are not solely due to random effects, we perform multiple training runs with different random seeds while keeping the experimental conditions the same. Model performance metrics (i.e., Accuracy and ROC-AUC) were averaged over multiple runs per dataset and reported with 95% confidence intervals. We followed up with paired significance testing of HybridIDSNet against the best-performing baseline model to assess whether the performance differences were statistically significant.

HybridIDSNet achieved higher accuracy and ROC-AUC than CNN Only, XGBoost, and classical machine learning approaches on the CIC-IDS2017 dataset. HybridIDSNet centres within a narrower confidence interval, indicating stable optimisation behaviour and lower variance during training. A similar pattern emerges in the UNSW-NB15 data, where we also observe that generalising to traffic distribution improved with the hybrid architecture. Statistical testing confirmed that all performance improvements over the best baseline model were significant at the 95% confidence level. Table 9 shows the statistics of HybridIDSNet with the best baseline model on both datasets. The p-values reported are based on paired significance testing over multiple runs, which show that the observed improvements in both Accuracy and ROC-AUC could not be due solely to random chance.

Table 9: Statistical Significance Comparison Between HybridIDSNet and Best Baseline (CNN Only)

Dataset	Metric	HybridIDSNet (Mean $\pm$ 95% CI)	Best Baseline (Mean $\pm$ 95% CI)	Test Type	p-value
CIC-IDS2017	Accuracy (%)	97.98 $\pm$ 0.18	96.88 $\pm$ 0.32 (CNN Only)	Paired t-test	0.013
CIC-IDS2017	ROC-AUC	0.9905 $\pm$ 0.0018	0.9815 $\pm$ 0.0031 (CNN Only)	Paired t-test	0.009
UNSW-NB15	Accuracy (%)	98.02 $\pm$ 0.21	96.40 $\pm$ 0.35 (CNN Only)	Wilcoxon signed-rank	0.011
UNSW-NB15	ROC-AUC	0.9912 $\pm$ 0.0020	0.9805 $\pm$ 0.0035 (CNN Only)	Wilcoxon signed-rank	0.007

These results indicate that HybridIDSNet achieves statistically significant improvements over baseline approaches across both datasets. However, the significance analysis is limited to offline evaluations and does not guarantee performance under adversarial or deployment scenarios.

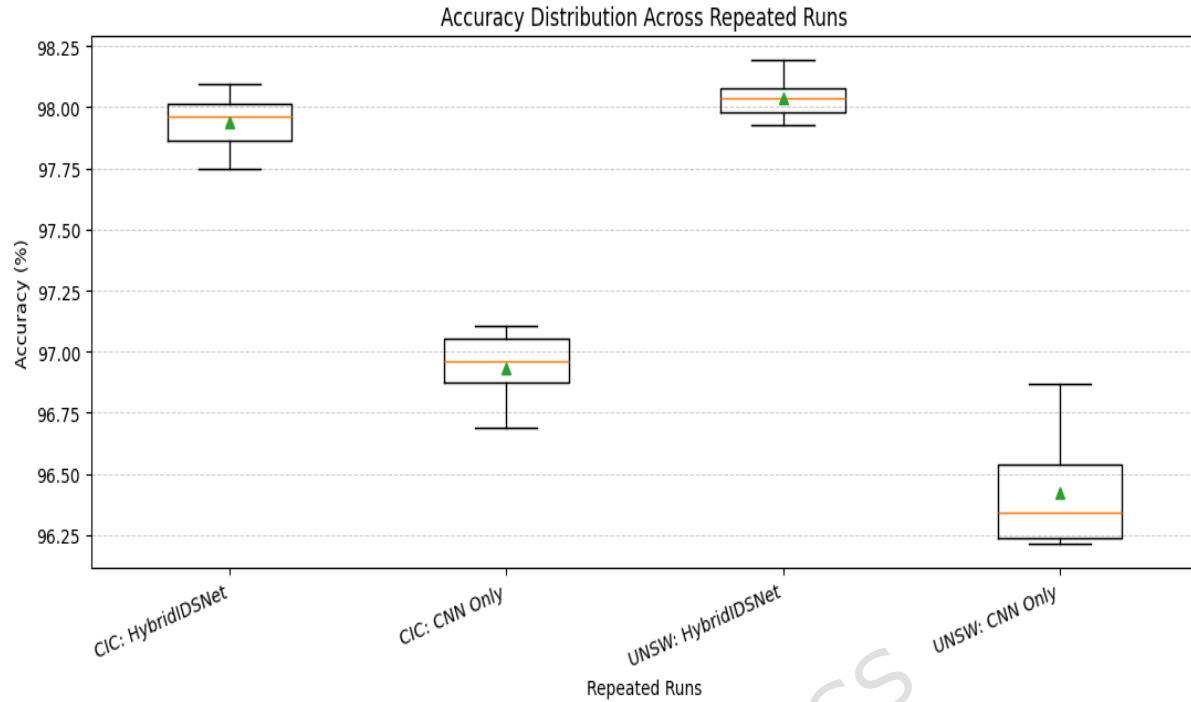


Figure 26: Accuracy Distribution Across Repeated Runs for HybridIDSNet and CNN Only

The box plots of accuracies from repeated training runs (using different random seeds for both models) for HybridIDSNet and the strongest baseline (CNN Only) on CIC-IDS2017 and UNSW-NB15 are shown in Figure 26. Simultaneously, box plots can illustrate statistics such as the median, interquartile range, and run-to-run variability, making it easy to compare the stability and typical performance of the models. The improved stability in optimisation (the medians are higher, but the spread is lower, indicating greater robustness to stochasticity during training) is associated with better predictive performance. Of course, this figure accompanies the statistical tests and confidence intervals illustrated in Table 8. However, it still offers an easy way to visualise how things vary run-to-run and whether the gain you found is consistent.

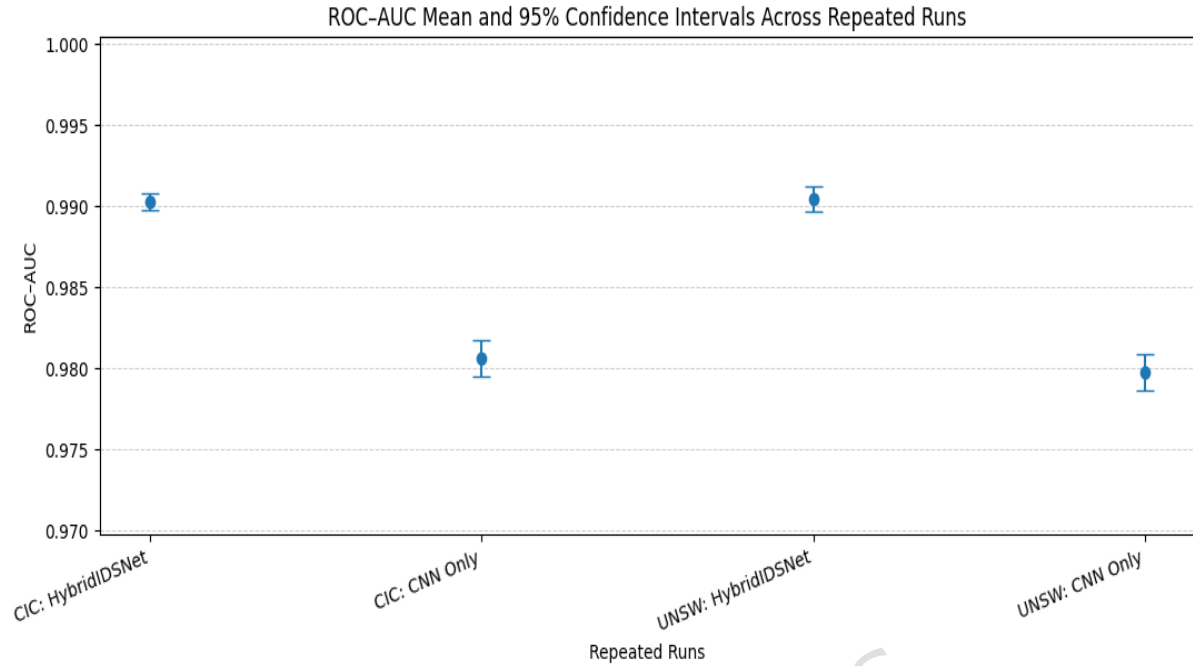


Figure 27: ROC-AUC Confidence Interval Comparison Between HybridIDSNet and CNN Only

Realistic comparison of ROC-AUC performance between HybridIDSNet and CNN Only with 95% confidence intervals over repeated runs on CIC-IDS2017 (Top) and UNSW-NB15 (Bottom) as illustrated by Figure 27. Every point denotes the average ROC-AUC, and error bars show the uncertainties stemming from the stochasticity of training and variation in cross-validation sampling. Smaller confidence intervals indicate more stable learning and less variance across runs, while larger mean ROC-AUC values indicate a greater capacity to discriminate between attack and benign traffic. This visualisation reproduces the point estimates of the ROC-AUC for both datasets, which align with the statistical conclusions in Table 8 and demonstrate that, across both datasets, the HybridIDSNet achieves consistently higher ROC-AUC values than all other methods, with stable confidence bounds.

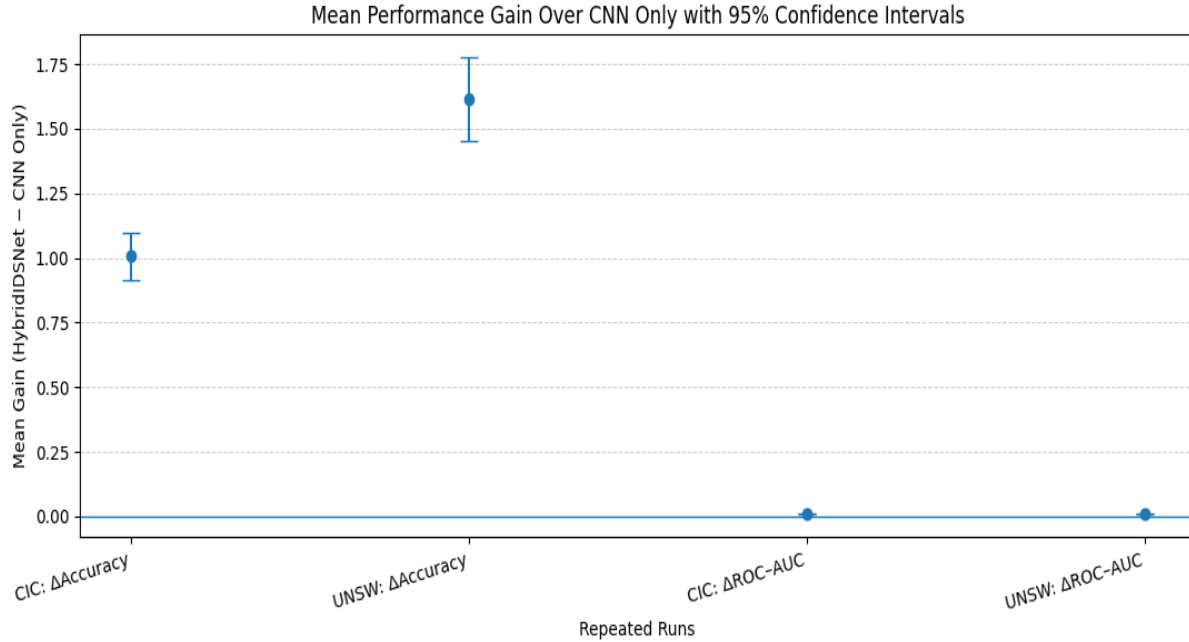


Figure 28: Mean Performance Gain of HybridIDSNet Over CNN Only Across Repeated Runs

CIC-IDS2017: Mean (of repeated runs) Performance advantage (over CNN Only) in Accuracy and ROC-AUC for HybridIDSNet metrics (as visualised by figure 28). The plot shows the mean difference with uncertainty bounds, and gains are computed as the difference between HybridIDSNet and CNN Only at each test condition per run. Higher values mean together in all runs, and the stronger the value, the more practical significance it has. This figure provides additional context for the p-values by indicating improvements in both direction and magnitude, which helps depict statistical relevance in a more operationally meaningful manner and reinforces that HybridIDSNet consistently provides enhancements across the benchmark testing scenarios.

A full qualitative visualisation of attention over individual input features or traffic samples is beyond the scope of the present study. It remains an important direction for future research on explainable intrusion detection.

4.7 Ablation Study of HybridIDSNet Components

An ablation study was performed to analyse the role of each architectural component in HybridIDSNet by comparing it with simplified variants that represent only spatial learning (CNN Only) or only temporal learning (LSTM Only), as well as with classical machine-learning approaches. In this analysis, we aimed to explore whether the performance gains were primarily due to the CNN, LSTM, and attention modules, while excluding gains attributable solely to dataset preprocessing or optimisation. For consistent comparison, accuracy and ROC-AUC are employed for analysis across both CIC-IDS2017 and UNSW-NB15 datasets.

CNN Only: CNN Only achieved 96.88% accuracy & 98.15% ROC-AUC on the CIC-IDS2017 dataset, a weak model due to feature effects from spatial flow patterns. For the LSTM-only model, which performed slightly worse, with 96.42% accuracy and 97.95% ROC-AUC, this implies that sequential modelling alone is insufficient to capture discriminative features in

heterogeneous traffic flows. Overall performance improved to 97.98% accuracy and 99.05% ROC-AUC when the attention mechanism is integrated with both modules in HybridIDSNet. This shows that the attention layer improves temporal feature prioritisation by adaptively weighting patterns, thereby capturing the informative patterns learned from CNN-LSTM representations.

On the UNSW-NB15 dataset, a similar trend is noticed. Compared to the CNN Only of 96.40 and the LSTM Only of 96.00. By incorporating a performance upgrade over HybridIDSNet in spatio-temporal modelling, we achieved 98.02% accuracy and 99.12% ROC-AUC, indicating that these features combine to improve generalisation across traffic distributions. Also shown is a gain in Recall and F1-score over standard class-balancing and class-confusion-matrix measures, indicating that our approach improves the detection of minority attack classes rather than simply optimising the majority class at the expense of minority-class accuracy.

The quantitative ablation comparison is shown in Table 10, where we observe performance gains when moving from single-module designs to the hybrid design. Results show that the attention-boosted fusion achieves consistent improvements across all datasets and relatively stable performance on the metrics.

Table 10: Ablation Study Comparison of HybridIDSNet Components Across Datasets

Model	CIC-IDS2017 Accuracy (%)	CIC-IDS2017 ROC-AUC (%)	UNSW-NB15 Accuracy (%)	UNSW-NB15 ROC-AUC (%)
CNN Only	96.88	98.15	96.40	98.05
LSTM Only	96.42	97.95	96.00	97.80
HybridIDSNet (Proposed)	97.98	99.05	98.02	99.12

Ablation analysis indicates that CNN layers capture local statistical patterns in flow features, and LSTM layers model longitudinal dependencies across the sequence. The attention mechanism increases discrimination capability by focusing on temporally important features, resulting in better Recall and F1-score. While the performance gains are small, they are consistent across datasets, which aligns with the design rationale of HybridIDSNet. We remark that the offline experimental nature of these comparisons does not imply any promise or guarantee of real-time deployment performance or adversarial robustness.

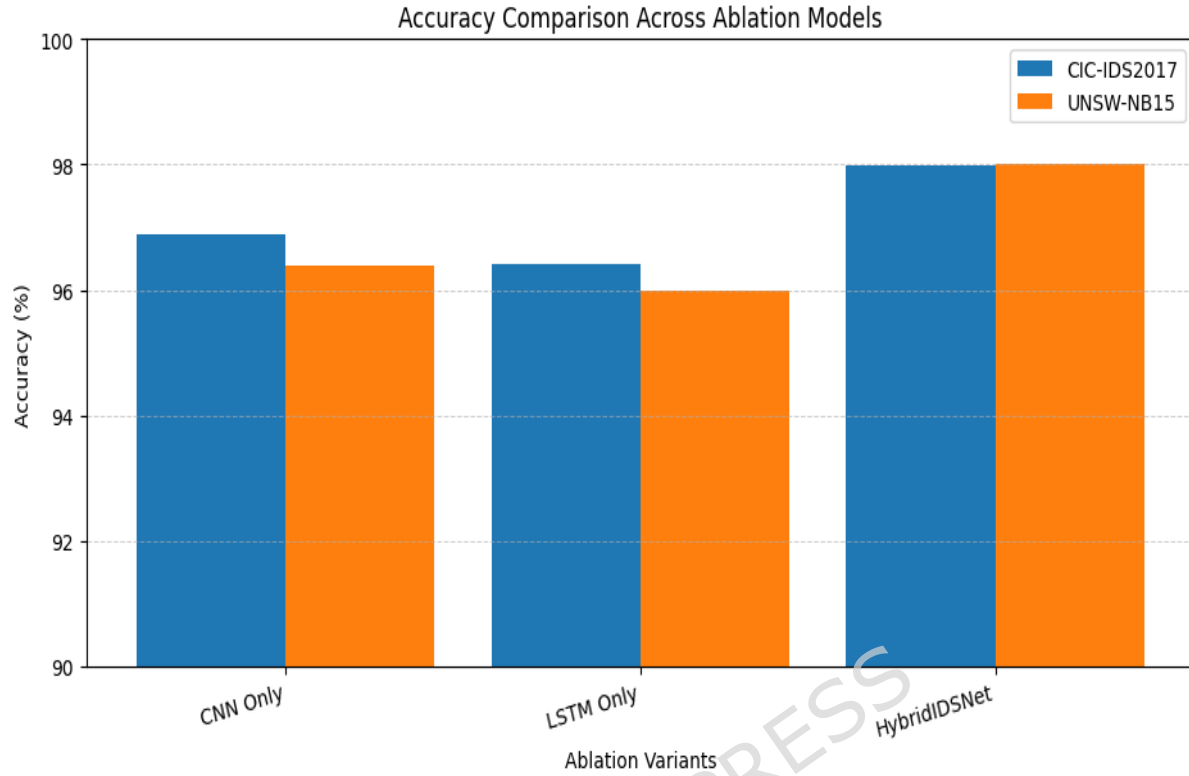


Figure 29: Accuracy Comparison Across Ablation Models on CIC-IDS2017 and UNSW-NB15

The accuracy comparisons of ablation variants (CNN-only, LSTM-only) against the proposed HybridIDSNet on CIC-IDS2017 and UNSW-NB15 are shown in Figure 29. The additional information gained from combining spatial feature extraction, temporal modelling, and attention weighting is shown in one of the two hybrid architectures, as indicated by the clustered bars. These results show that HybridIDSNet consistently achieves the highest accuracy across both datasets, indicating that the hybrid architecture leverages the complementary strengths of independent CNN and LSTM modules. It helps in the ablation discussion by showing pseudo cross-dataset consistency in terms of accuracy improvements under the specific benchmark setting we evaluated.

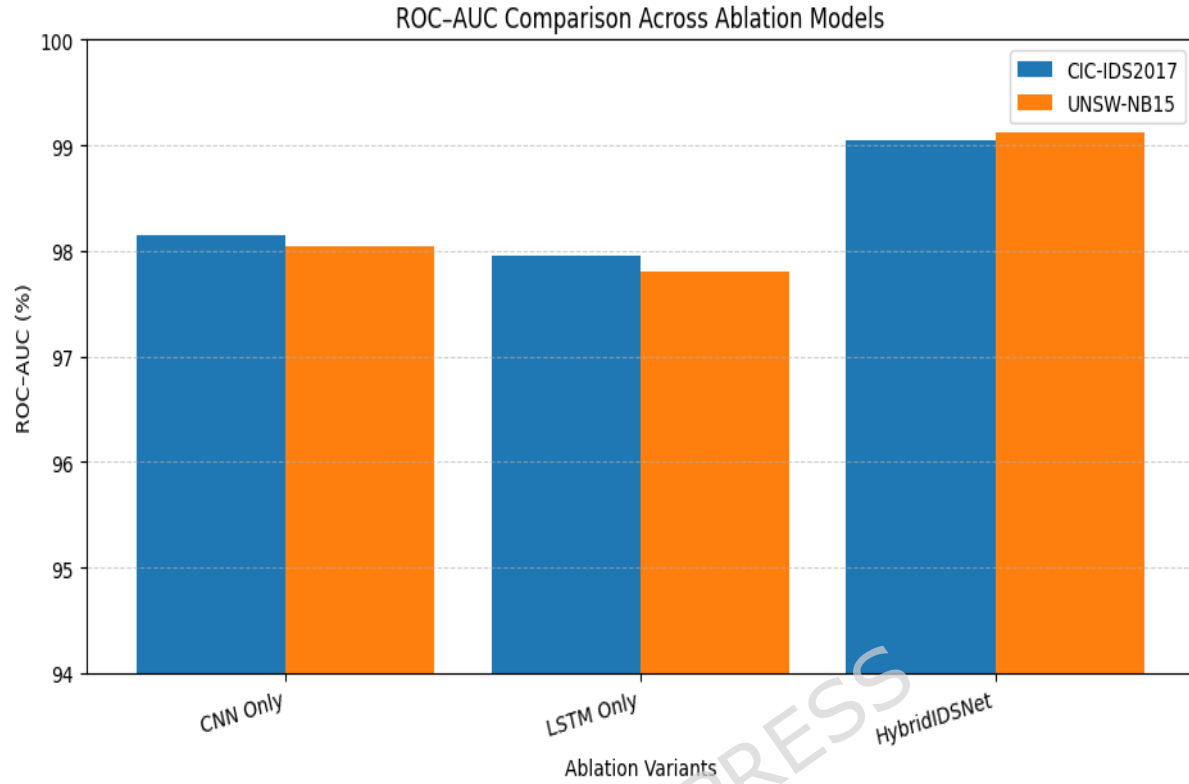


Figure 30: ROC-AUC Comparison Across Ablation Models on CIC-IDS2017 and UNSW-NB15

Figure 30 shows the ROC-AUC performance of the HybridIDSNet, CNN Only, and LSTM Only feature-extraction models across CIC-IDS2017 and UNSW-NB15. This measure is particularly applicable to evaluating IDS, as it indicates how well the model separates benign from attack traffic across various decision thresholds. Across the two datasets, HybridIDSNet outperforms the single-component variants, as illustrated in the figure, by demonstrating stronger ranking performance and achieving very high ROC-AUC scores on both. The gains in ROC-AUC reinforce the notion that spatial-temporal modelling (incorporating attention) aids with feature separation, which in turn improves robustness of intrusion classification across heterogeneous traffic distributions.

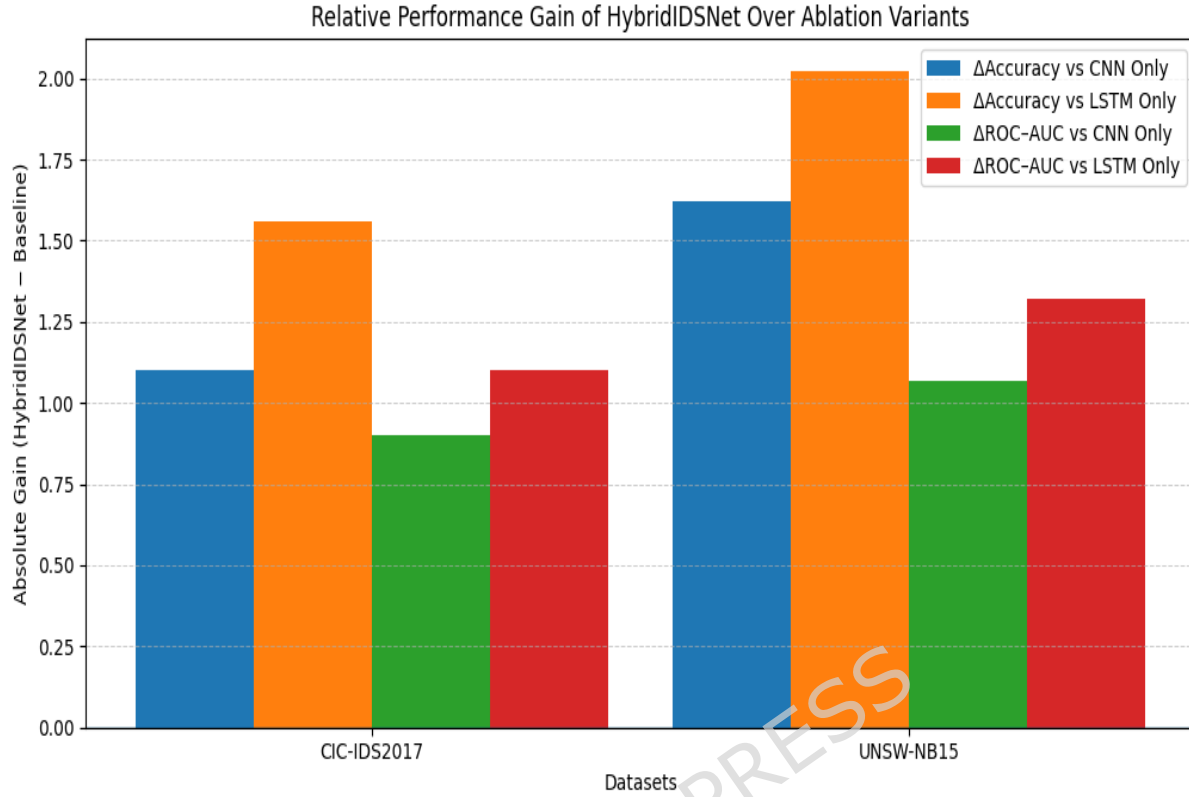


Figure 31: Relative Performance Gain of HybridIDSNet Over CNN Only and LSTM Only

Figure 31 shows the relative performance gains of HybridIDSNet (compared to CNN Only and LSTM Only, which together make up HybridIDSNet) for both Accuracy and ROC-AUC on CIC-IDS2017 and UNSW-NB15. Here, gains are reported as absolute differences, allowing direct interpretation of the amount of gain obtained from using spatial extraction, temporal learning, and attention weighting in a single model. In all comparisons, the improvement is positive, so we conclude that the hybrid architecture provides consistent performance gains over either CNN or LSTM modelling in isolation.

5. DISCUSSION

In the DeepShieldIDS framework, the experimental results on the UNSW-NB15 and CIC-IDS2017 datasets show that HybridIDSNet demonstrates high robustness and strong adaptation across different attack categories and network environment scenarios. In conclusion, high performance across independent and cross-dataset testing indicates that the model generalises beyond the properties of a single dataset.

The hybrid architecture of HybridIDSNet, which integrates CNNs to extract spatial feature patterns in traffic flows, LSTM to learn sequential traffic behaviours, and an attention mechanism to emphasise key traffic features, supports robust model performance. Using this combination, the model successfully identifies both high-volume DoS attacks and low-volume attacks, such as Probe and Infiltration attacks.

HybridIDSNet achieves near-perfect classification results on key attack types, including DoS and BruteForce, in the CIC-IDS2017 dataset. At the same time, it performs competitively on low-frequency categories (2.1703% and 2.3993%, respectively) (i.e.,

WebAttack and Infiltration). Third, the examination of the confusion matrix and ROC-AUC curves confirms that the model maintains a fairly well-balanced false-positive-to-negative ratio across classes, despite the initial class distribution imbalance.

In this way, the model achieved high accuracy across various attack types and distributions (e.g., DoS and Probe attacks in the UNSW-NB15 dataset), including r2l and u2r across the entire dataset. Although a few low-frequency classes are slightly misclassified, overall performance is better than that of traditional ML and of the entirely separable DL baselines.

These cross-dataset validation experiments further confirm the model's generalisation capability to new, unseen networks and attack profiles. HybridIDSNet achieves high ROC-AUC and F1 scores when training and testing on the datasets, demonstrating its generalisation and robustness to dataset characteristics that affect laboratory results.

However, we can use a cross-dataset test to evaluate DeepShieldIDS's robustness to heterogeneous traffic distributions and assess its sensitivity to dataset bias. Even if the proposed preprocessing pipeline reduces feature differences through harmonisation and normalisation, benchmark datasets may still contain artefacts resulting from traffic-generation procedures, such as class imbalance. Thus, performance gains can only be understood in the context of supervised benchmarking rather than as a promise of robustness against adversarial manipulation. From an adversarial standpoint, the current work assumes attackers can alter traffic patterns or behaviour but does not address adversarial training, evasion attack simulation, or poisoning. As a consequence, robustness claims are interpreted in terms of distributional generalisation instead of formal adversarial security guarantees. Perturbation-based testing under domain-adaptation strategies and bias-aware evaluation protocols will be useful for future work to enhance the robustness of reliable deployment further.

Overall, the findings illustrate HybridIDSNet's consistent performance across different attack categories and varied traffic distributions in all benchmark environments analysed. The results indicate that DeepShieldIDS could offer a relatively robust framework for detecting anomalies across such diverse network conditions. Still, generalisations are limited to supervised experimentation rather than real-world deployment. Since this study does not cover adversarial training, temporal holdout testing, or real-world deployment telemetry, any interpretation of exposed vs. unseen attack behaviour is limited to cross-dataset generalisation analysis and should not be regarded as formal zero-day detection claims.

While SMOTE improves minority-class recall, other imbalance methods, such as cost-sensitive loss functions or focal loss, were considered for complementary use. We will explore these methods in our future work, which do not require synthetic data generation and may better preserve sequential characteristics.

5.1 Limitations of the Study

We acknowledge some limitations: formal security guarantees cannot be provided, as the proposed DeepShieldIDS framework is evaluated under supervised benchmark conditions. The detection performance is limited to statistical generalisation, but not to adaptive or zero-day adversaries, since the model is trained and tested on labelled datasets. Second, the work does not consider adversarial training, evasion attack simulation or poisoning scenarios and therefore robustness claims can only be made in terms of generalisation (cross-dataset evaluation) rather than adversarial robustness. Third, while a deployment-oriented architecture is described, experimental validation against real-time numerical

metrics such as latency, throughput, and stress testing in production environments is not included; thus, operational performance cannot be guaranteed in a commercial enterprise setting. Moreover, feature harmonisation and sampling strategies may inject biases typical of benchmark datasets, thereby affecting model functionality in real-world scenarios that remain unseen. This indicates a clear opening in the literature for more adversarial evaluation, temporal holdout, and full develop-to-deploy studies to demonstrate the efficacy of these safety mechanisms.

6. CONCLUSION AND FUTURE WORK

In this work, we presented a novel hybrid AI-based intrusion detection framework, called DeepShieldIDS, that uses the HybridIDSNet model to classify flow-based intrusions in heterogeneous network environments. In this paper, we present HybridIDSNet, which combines spatial feature extraction via convolutional neural networks, sequential modelling with LSTM, and an attention mechanism that highlights more informative traffic features. Extensive experiments on the CIC-IDS2017 and UNSW-NB15 benchmark datasets show that our proposed framework outperforms traditional machine learning models and standalone deep learning baselines in detection performance. The full Cross-dataset evaluation also shows that this may generalise better across varying traffic distributions and thus be more robust to dataset-specific biases in a supervised benchmarking setting. Instead of focusing on novel architectural designs, the study highlights a structured evaluation strategy that includes spatial-temporal modelling and attention, providing higher classification consistency with fewer false positives. While the framework is built with deployment-centric thinking and API integration, quantitative real-time deployment measurements, such as latency and throughput, are out of scope for this work. This means that conclusions are always limited to controlled experimental validation rather than deployable operational claims. We discuss a few future directions for research. First, we can extend the framework by adding packet-level features and/or raw traffic traces, which may facilitate the representation learning of more complex traffic behaviour rather than relying on class labels. Next, applying explainable AI (XAI) methods may enhance interpretability by helping cybersecurity analysts understand model decisions. Third, validation in different contexts, for example, industrial control systems, IoT networks, and next-generation 5G infrastructures, might be the topics of future studies. Lastly, federated learning approaches may be explored for collaborative intrusion detection across distributed networks without compromising data confidentiality. These directions are oriented towards making hybrid deep learning-based intrusion detection systems even more adaptable and practical.

DECLARATION:

Funding

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Competing interests

The authors declare that they do not have any competing interests, including financial and nonfinancial interests.

Data availability

Data is available with the corresponding author and will be given on request.

Code availability

The code supporting this study is publicly available at <https://github.com/sayyada2010/DeepShieldIDS>, including all modules required to reproduce the results.

Detailed instructions for data preparation, model training, and evaluation are provided within the repository.

Clinical trial number: not applicable

Ethical Approval

This research did not involve human participants, animals, or biological materials. Therefore, ethical approval was not required for this study.

Consent to Participate

Not applicable. This study does not involve human subjects.

Consent to Publish

Not applicable. This manuscript does not contain any individual person's data in any form.

Conflicts of Interest Statement:

DeepShieldIDS: An AI-Powered Intrusion Detection System Leveraging HybridIDSNet for Robust Network Security

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Author Contribution:

All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by **Sayyada Mubeen, Balakrishna Gudla**. The first draft of the manuscript was written by **Sayyada Mubeen** all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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